

USER GUIDE

**Develop. Mask.
Measure. Deliver.**

A complete path through organizing a session, developing RAW files, building local corrections, and preparing repeatable output.

RAW workflow Sessions and export	Masks SCS, brush, and AI	Geometry Optics and perspective
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VERSION 0.6.5 · ENGLISH EDITION

ChromaLux User Guide

Documented version: ChromaLux 0.6.5

Language: English

Editorial status: 0.6.5 edition

Validation of this page: navigation and local links checked; the English mirror has not been replayed page by page in the final package

ChromaLux organizes photographs into sessions and stores adjustments non-destructively. This guide starts with the essential workflow: create a session, import photographs, develop a RAW, and export it.

The French guide remains the editorial source of truth. The internal docs/user/ PARITY_FR_EN.md matrix states exactly which English pages are complete, condensed, or waiting for a product replay. Terminology is fixed in the internal docs/user/GLOSSARY_FR_EN.md glossary. These editorial control files are not chapters of the distributed guide.

Get started

- [Quick start](#) — produce a first image in about fifteen minutes.
- [Tour the interface](#) — understand activities, modules, the Viewer, and the Browser.
- [Formats and compatibility](#) — check supported cameras, files, profiles, and outputs.
- [Initial troubleshooting](#) — solve common issues without risking originals.

Understand ChromaLux

- [Sessions and files](#) — locate the catalogue, originals, and sidecars.
- [Selection and nondestructive editing](#) — distinguish the focused photo, selection, variants, and batch operations.
- [Color management](#) — follow color from the SCSP camera profile to the Viewer, display, delivery, and proof profiles.

- [Activities, tabs, and modules](#) — organize the workspace and identify incomplete features.
- [Keyboard shortcuts](#) — find the default commands and their scope.
- [Accessibility and assisted navigation](#) — use keyboard focus, descriptions, density controls, and the documented accessibility limits.

Production workflows

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Studio and advanced tools

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Topics still to expand

This edition covers primary workflows and 0.6.5 Studio tools. Future chapters can add a parameter-by-parameter reference, HALO and SCS principles, a broader camera/lens/format qualification matrix, and Web publication from the same corpus as the offline guide.

How to read this guide

Names in **bold** match buttons, activities, or menus in the English ChromaLux interface. A path such as **File > Import Images...** lists controls to open in sequence.

Product maturity and documentation validation are separate. A feature can be **Stable** while its documented procedure still requires a replay. The English pages preserve the French validation statement; translating a page does not upgrade its validation.

ChromaLux 0.6.5 compatibility

The current package targets Apple Silicon Macs and requires macOS 26 Tahoe because of the bundled components. Check [formats and compatibility](#) before production work with a new camera or unusual format.

Quick start

Documented version: ChromaLux 0.6.5

Product status: Stable

Documentation validation: PARTIAL

Evidence: launch, session, import, and interface replayed on a 0.6.5 package predating this documentation revision

Blocker: development, export, and final-package replay to be completed

English mirror: NOT REPLAYED

Estimated time: 15 minutes

Outcome: one imported, adjusted, and exported photograph

This path uses the defaults and quickly checks installation, session handling, the Viewer, and export.

1. Install and launch ChromaLux

1. Open the ChromaLux 0.6.5 DMG.
2. Copy **ChromaLux** to **Applications**.
3. Launch it from **Applications**.

Version 0.6.5 requires an Apple Silicon Mac running macOS 26 Tahoe.

To display the French interface, choose **View > Language > French**, then restart ChromaLux. The language change applies at the next launch.

2. Create a session

At first launch, ChromaLux asks you to create a session. A session is a working folder containing a `.cls` session catalogue and folders for captures, selections, output, and items moved to the session trash.

1. Enter a clear session name, such as `Client_Portrait_July`.
2. Choose the parent folder with **Browse....**

3. If needed, enter the photographer, client or project, campaign or series, and job name.
4. Keep the proposed folder names for this first session.
5. Confirm creation.

The **Add capture folders** field contains this default expression:

```
/00010 +10 12
```

It creates twelve folders, from 00010 through 00120, in increments of ten. Clear the option if you do not need that structure.

Nouvelle session

Nom : Session Demo

Emplacement : /Users/Shared Parcourir...

Identité de la session

Photographe : Photographe ou s...

Client / Projet : Client ou projet

Campagne / Série : Campagne ou série

Nom du budget : Nom du budget o...

Ajouter un champ...

Réglages des dossiers

Sous-dossier Captures : Captures

Sous-dossier Sélections : Sélections

Sous-dossier Sortie : Sortie

Sous-dossier Corbeille : Corbeille

Nom de capture : Session Demo

☒ Ajouter des dossiers de capture

/00010 +10 12

Un dossier par ligne. Utilisez « / » pour les sous-dossiers, ou « /00010 +10 12 » pour créer 12 dossiers de 00010 à 00120.

Annuler OK

French ChromaLux 0.6.5 capture supplied by the source guide. No English UI capture is claimed.

After creation, ChromaLux opens the **Library** activity and selects the session capture folder.

3. Import a few photographs

1. Choose **File > Import Images...** or press ⌘I.
2. Under **Source**, choose the folder or volume containing the photographs.
3. Under **Destination**, choose the session capture folder.
4. Check the thumbnails and optional settings.
5. Click **Import**.

For this first test, leave the card intact and keep **Erase images after copying** disabled. In 0.6.5 this option is available only with **Source + destination checksum** verification and **Emit ASC-MHL**. It permanently deletes imported originals from the source only after a completely successful batch and explicit confirmations. Reserve it for a workflow you have already qualified.

4. Display a photograph

Click one thumbnail in the **Browser**. The current photograph appears in the central Viewer. Use ← and → for the previous or next image.

The first RAW display can take a moment while it is decoded. The information bar below the Viewer shows capture data and zoom.

Warning — selection synchronization is enabled by default. Keep only one photograph selected for this first development so that you do not unintentionally apply an adjustment to a batch.

5. Make a simple development

Open **Adjustments**, then work in this order:

1. In **RAW Interpretation**, check the camera profile and RAW demosaicing.
2. In **White Balance**, correct an overly warm or cold cast.
3. In **Exposure**, set overall brightness, then contrast if needed.

4. In **Extended Dynamic Range**, recover highlights or shadows moderately.
5. In **Detail**, adjust sharpening and noise reduction at a sufficiently high zoom.

Press / to compare temporarily with the rendered RAW without your adjustments.

The **BEFORE** badge means user adjustments are bypassed. Press / again to return to the development.

Adjustments are nondestructive: they do not replace the original photograph.

6. Export the result

1. Open **Export**.
2. Check at least one recipe under **Recipes**.
3. Select it and check its format, location, and naming template.
4. Choose a destination. For a first test, use the session **Output** folder.
5. Click **Export**.

The button remains disabled until a photograph is loaded and at least one recipe is checked.

The resolution field writes resolution metadata; it does not by itself change pixel dimensions. Use the recipe scale to resize the image.

Check the result

Open the destination from the export panel or macOS Finder, then check:

- file name and format;
- dimensions;
- color and brightness in a color-managed application.

Continue with [Import, develop, and export](#) and [Tour the interface](#).

Tour the interface

Documented version: ChromaLux 0.6.5

Product status: Stable, with incomplete functions identified

Documentation validation: PARTIAL

Evidence: targeted visual check on an earlier 0.6.5 package and audit of activities, modules, preferences, and translations

Blocker: new density, color, and accessibility states not replayed in the final package

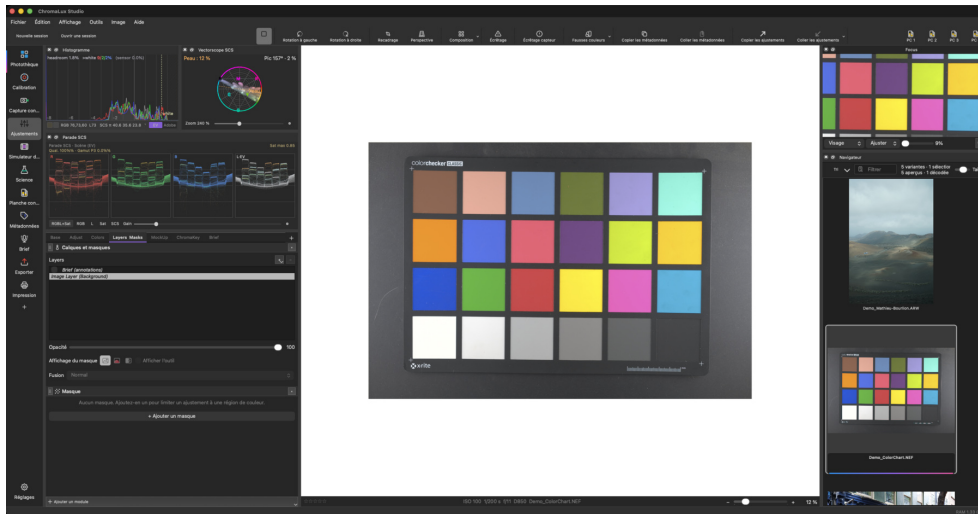
English mirror: NOT REPLAYED

Outcome: know where to find photographs, tools, and outputs

ChromaLux is organized around a central image and modular panels. Changing activity adapts the tools without changing the selected photograph.

Main areas

Area	Purpose
Activity Bar	Changes the production stage: Library, Adjustments, Export, and others.
Left panels	Contain modules for the current activity.
Central Viewer	Displays the current photograph and visual tools.
Focus	Magnifies an area for sharpness inspection.
Browser	Displays thumbnails and selects one or more photographs.
Toolbar	Provides frequent commands and can be customized.
Viewer Info Bar	Shows capture data, rating, labels, and zoom.



The source guide supplies this French 0.6.5 capture. It is not an English UI replay.

Activities

Activity	Main use
Library	Browse session folders and filter photographs.
Calibration	Measure a CC24 chart and manage profiles. Studio only; Scientific , Detector , and Geometric still announce future phases.
Tethered Capture	Operate a compatible connected camera. Studio only.
Adjustments	Develop RAW files and grade images.
Film Simulator	Add film curve, grain, halation, black and white, or a darkroom-print rendering.
Science	Use specialized measurement and visualization tools.
Contact Sheet	Compose a selection and its caption.

Metadata	Read and edit photograph and session information.
Brief	Annotate an image for production exchanges.
Export	Produce files with one or more recipes.
Print	Prepare printed output.

Some activities can be hidden by the edition, settings, or enabled features.

Panels, tabs, and modules

Default **Adjustments** modules are grouped as follows:

- **Base:** RAW Interpretation, White Balance, Exposure, HDR, Detail, and Primaries;
- **Adjustments:** curves, levels, tone equalizer, dehaze, fringes, and vignetting;
- **Colors:** Color Balance, Color Editor, and Harmony;
- **Layers & Masks:** localized adjustments;
- **MockUp:** RAW mockup preparation and composition;
- **Brief:** annotations when enabled.

Module headers collapse modules. Modules and panels can be reordered, grouped into tabs, detached, or removed. **Tools** opens the module catalogue. Add the **Module** command to the toolbar if you want direct access.

If the layout becomes hard to recover, choose **View > Reset Default Layout**. This restores panels without deleting photographs or adjustments.

Viewer and Browser

The Viewer displays the focused photograph. The Browser can hold a multiple selection, but exactly one photograph is focused for tools that use the current image.

- Click a thumbnail to focus it.
- Use ← and → to navigate.
- Use the Viewer zoom control to fit the image or inspect details.

- Choose **View > Open Focus Loupe** for high magnification.
- Choose **View > Open Second Viewer** for another window or display.

Copy commands take adjustments or metadata from the focused photograph. Paste then applies them to the selection defined by the command.

Warning — selection synchronization is enabled by default. When enabled, an adjustment can propagate to other selected photographs. Check the Browser selection before an individual edit.

Toolbar

The top toolbar includes session creation/opening, margins, rotation, crop, perspective, composition guides, clipping, metadata copy, adjustment copy, and four contact-sheet viewers. Text appears below icons in the default layout.

Perspective is distinct from **Crop**. It opens vertical and horizontal guides without changing the crop frame. See [Correct lens distortion and perspective](#).

To customize the toolbar:

1. right-click it or choose **View > Customize Toolbar...**;
2. add, remove, or reorder commands;
3. use **Reset Toolbar** when needed.

Interface and color preferences

Open preferences with `⌘,`.

Under **General > Appearance**:

- choose light, dark, or system theme;
- choose **Compact** or **Comfortable** density; Comfortable keeps labels below Activity Bar icons;
- enable **Neutral color environment** to replace decorative colors with achromatic greys while judging color;
- ChromaLux refuses a custom surface palette whose text contrast is below 4.5:1.

Under **Viewer > Color**, **Viewer render space** selects sRGB, Adobe RGB (1998), or

Display P3. It does not select an export recipe ICC. **Active display profile** identifies the ColorSync profile of the display carrying the window, with SHA-256 in its tooltip.

See [Color management](#) before changing these boundaries.

Language, shortcuts, and accessibility

View > Language requires a restart. Shortcuts are visible and editable in **View > Keyboard Shortcuts....**

Action	Default shortcut
New session	⌘N
Open session	⌘O
Import images	⌘I
Undo / redo	⌘Z / ⇧⌘Z
Previous / next image	← / →
Show RAW without adjustments	/
Activate crop	C
Activate hand tool	H

Import, Browser, layers, masks, and export queue expose accessible names and descriptions. VoiceOver has not been replayed in the final package. See [Accessibility and assisted navigation](#).

Sessions and files

Documented version: ChromaLux 0.6.5

Product status: Stable

Documentation validation: NOT REPLAYED

Evidence: audit of code, translations, and associated-file formats

Blocker: creation, closing, and reopening not replayed in the final package

English mirror: NOT REPLAYED

Outcome: understand where the catalogue, originals, metadata, and adjustments are stored

A ChromaLux session is an ordinary working folder. Originals remain visible files on disk; the catalogue does not enclose them.

Session structure

A session named July_Portrait has this default structure:

```
July_Portrait/  
├─ July_Portrait.clssession  
├─ Captures/  
├─ Selects/  
├─ Output/  
├─ Trash/  
├─ Styles/  
├─ Workflows/  
└─ .cls/
```

Item	Purpose
July_Portrait.clssession	Catalogue paths, search index, ratings, colors, statuses, and session data.

Captures	Default destination for imports and tethered captures.
Selects	Folder intended for selected photographs.
Output	Default export destination.
Trash	Session folder for photographs removed from the current workflow.
Styles, Workflows	Reusable styles and workflows.
.cls	Internal data, including locks, logs, and the AI metadata cache.

The four role-folder names can be changed in **New Session**. Extra folders entered under **Add capture folders** are created beneath the chosen capture folder. For example, `/00010 +10 12` creates `00010` through `00120`.

Catalogue and photographs have different roles

The `.cls` session file indexes photographs and stores session identity: photographer, client or project, campaign or series, job name, and production team. Photographs remain separate RAW, JPEG, or HEIC files.

Deleting the catalogue does not alter originals, but loses catalogue-only organization and information. Edit session identity later in **Metadata**. Changes feed future naming, import, tether, metadata, and contact-sheet operations; they do not rewrite files already processed.

Files associated with a photograph

For `image.nef`, ChromaLux can use:

`image.nef`

`image.nef.xmp`

`image.nef.cls.xmp`

`ChromaLux/previews/`

Item	Main content
<code>image.nef</code>	Photographic original.
<code>image.nef.xmp</code>	Descriptive metadata, rating, and color label.
<code>image.nef.cls.xmp</code>	ChromaLux development: adjustments, crop, variants, layers, measurements, and application data.
ChromaLux/previews/	JXL previews, including distinct thumbnails per variant.

The two XML sidecars have different jobs: `.xmp` carries descriptive metadata; `.cls.xmp` carries ChromaLux processing and variants.

Warning — keep a photograph with both sidecars. Without `.cls.xmp`, ChromaLux cannot restore its saved development and variants.

Move or copy photographs

Dragging from the Browser to ChromaLux Finder moves a photograph and its sidecars together. Hold $\mathbb{W}$ while dropping to copy instead. ChromaLux updates the catalogue.

Moving files directly in macOS Finder does not update the catalogue and can separate a sidecar from its original.

To move a complete session:

1. quit ChromaLux so pending writes finish;
2. copy the complete session folder;
3. open its `.clssession` file from the new location;
4. check several photographs and adjustments before deleting the old copy.

Relative paths preserve the organization of photographs inside the session.

Browse outside the session

ChromaLux Finder can show a folder outside the active session. It does not automatically copy those files into **Captures**. If you edit such a photograph, its

sidecars are stored beside the original.

Import or copy originals into the session before the main work if the production must be portable and straightforward to back up.

Back up a session

Quit ChromaLux, then back up the complete folder. It preserves:

- the `.cls` session catalogue;
- originals located inside the session;
- descriptive and ChromaLux sidecars;
- outputs, styles, and internal data.

Verify the copy on a second volume before formatting a card or deleting an earlier source. A catalogue-only backup contains no photographs.

See [Selection and nondestructive editing](#) and [Import, develop, and export](#).

Selection and nondestructive editing

Documented version: ChromaLux 0.6.5

Product status: Stable

Documentation validation: NOT REPLAYED

Evidence: audit of code, translations, and selection tests

Blocker: complete procedure not replayed in the final package

English mirror: NOT REPLAYED

Outcome: know which photograph or variant receives an adjustment without changing the original

ChromaLux separates the original, its development variants, the Browser selection, and the focused photograph.

Four distinct concepts

Concept	Meaning
Original	Photograph file on disk. Adjustments do not replace its pixels.
Variant	Independent development attached to the same original.
Selection	Set of thumbnails a batch command can target. It can contain two variants of one original.
Focused photo	Active thumbnail inside the selection. It feeds panels and is the source for copy commands.

In a multiple selection, the active outline identifies the focused photograph or variant. Clicking another already-selected thumbnail changes focus without discarding the selection.

Build a selection

In the Browser:

- click an unselected thumbnail without a modifier to replace the selection;
- ⌘-click to add or remove separate thumbnails;
- ⇧-click to select a continuous range;
- ⌘⇧-click to add a range without removing other groups;
- with multiple thumbnails selected, use ← and → to move focus without reducing the selection.

Clicking empty grey space clears the selection but leaves the last image in the Viewer.

Multiple selection and split view

A multiple selection opens a split view with up to twelve photographs or variants. The focused tile has a distinct outline and adjustment panels show its values.

If more than twelve items are selected, only twelve Viewer slots are shown. The full selection remains available to batch operations, including adjustment application, export, and print.

Two variants of one original can appear side by side. They share the source file but retain separate adjustments and thumbnails.

Choose adjustment scope

The split-view bar contains **Sync All**. It is enabled by default in a new 0.6.5 setup; ChromaLux then remembers the last setting.

State	Effect of a global adjustment
Sync All enabled	A change made from the focused photo applies to every selected photo or variant.
Sync All disabled	Only the focused photo or variant receives the change.

Only the changed setting is propagated; other existing adjustments on each destination are preserved.

Warning — check the selection and **Sync All** before moving a slider. The button is

visible only when several items are displayed.

A selected local layer remains specific to the focused photograph even when **Sync All** is enabled. Select the background/global adjustment to resume selection synchronization.

Create and use variants

Right-click a thumbnail and open **Variant**:

- **New Variant** creates a neutral development;
- **Clone Variant** copies the chosen development and layers;
- **Delete Variant** removes that development without deleting the original.

ChromaLux always keeps at least one variant. Each variant has its own Browser entry; check its name and active outline before editing.

Copy a development

Copy Adjustments always reads from the focused photograph or variant:

1. focus the source;
2. choose **Copy Adjustments**;
3. build the destination selection;
4. check **Sync All**;
5. choose **Paste Adjustments**.

With **Sync All** enabled, paste targets the full selection. Otherwise, it targets only the focused item. The paste menu can limit transferred adjustment groups.

Meaning of “nondestructive”

Exposure, color, crop, and other settings are stored in `image.ext.cls.xmp` beside the original. The source RAW, JPEG, or HEIC is not rewritten.

This makes it possible to:

- revise an adjustment without degrading the original;
- keep multiple variants of one file;
- compare with the adjustment-bypassed rendering using `/`;

- create a new file only at export.

BEFORE is temporary and does not delete the sidecar.

Warning — nondestructive does not mean “no working files.” Deleting `.cls.xmp` loses saved adjustments and variants even though the original remains intact.

Three safe methods

Correct one photograph

1. reduce the selection to one thumbnail or disable **Sync All**;
2. check the focused variant;
3. adjust it;
4. inspect the thumbnail and Viewer.

Adjust a coherent batch

1. select photographs made under similar conditions;
2. focus a representative image;
3. enable **Sync All**;
4. make small corrections;
5. inspect several selected images.

Compare two interpretations

1. clone the current variant;
2. edit the clone;
3. ⌘-click both variants;
4. compare them in split view;
5. disable **Sync All** before differentiating them further.

See [Tour the interface](#) and [Sessions and files](#).

Color management

Documented version: ChromaLux 0.6.5

Product status: strict v2/v3 pipelines, bounded SCSP v3 DCP support, and EDR preview for v3; external profiles and devices must be qualified for each workflow

Documentation validation: NOT REPLAYED

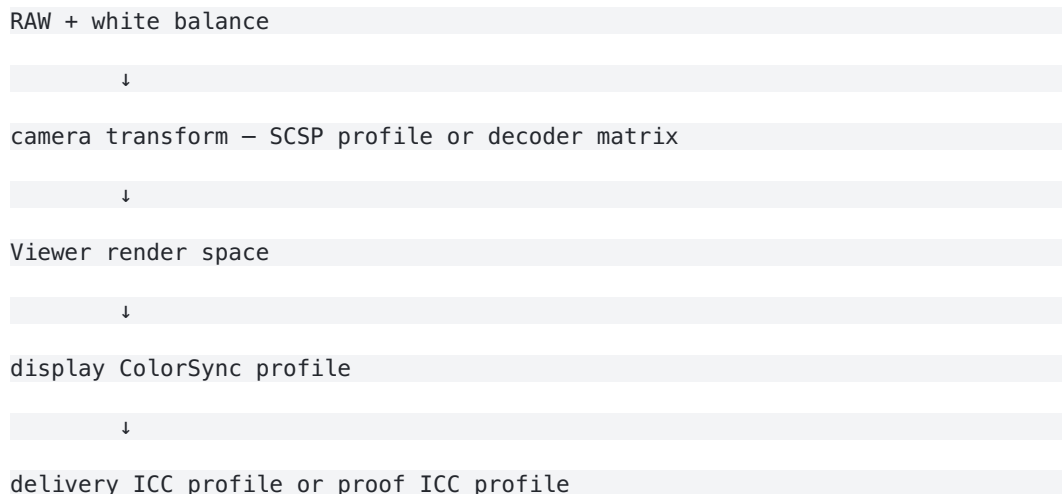
Evidence: audit of code and automated profile, rendering, soft-proofing, and export tests

Blocker: no final-package replay with a calibrated display, a real print profile, and a measuring instrument

English mirror: NOT REPLAYED

Outcome: distinguish camera, Viewer, display, delivery, and proof profiles without applying one conversion twice

ChromaLux does not have one all-purpose “color profile.” Four successive boundaries have different responsibilities:



Do not substitute one profile for another because their names look similar. SCSP describes camera input; a display ICC profile describes a monitor; a recipe ICC profile describes the delivered file or device.

Two independent versions

ChromaLux versions a variant's rendering pipeline separately from its camera profile file:

Version	Scope	Behavior
color_pipeline_version v1	topology and pixel handling of a historical variant	frozen pipeline for reproducing the old rendering
color_pipeline_version v2	strict SDR variant	explicit color inputs, high-precision linear transport, terminal display transform
color_pipeline_version v3	current new variant	v2 topology plus the versioned EDR-preview contract
SCSP v1	historical camera-profile file	readable but classified legacy ambiguous
SCSP v2	strict camera-profile file	mandatory domains, headroom, illuminant, provenance, and hashes
SCSP v3	strict dual-illuminant DCP file	exact validated DCP payload, two illuminants, runtime solve, and hashed provenance

A variant without `color_pipeline_version` is interpreted as v1. A newly created variant receives v3; a normal clone preserves its source version. Unknown values are refused. ChromaLux never silently migrates v1 or v2 because their pixels could change. **Duplicate active variant for HDR** is the explicit, non-destructive operation that creates a v3 copy while retaining the source.

In pipelines v2 and v3, an already-rendered image with ICC is converted to the high-precision linear working domain. Without ICC it is explicitly assumed sRGB; a ChromaLux-rendered DNG identified as linear keeps that domain. Pipeline v1 retains

its historical 8-bit route for these rendered inputs.

Reading SCSP v1 likewise never turns it into SCSP v2. The two compatibility mechanisms are independent: a pipeline version does not establish the quality of the selected profile contract.

1. Characterize the camera with SCSP

The camera profile is applied after RAW decoding and white balance. It maps the camera's RGB into ChromaLux's linear working domain. This stage remains scene-referred, so values above nominal white can retain highlight headroom.

Choose the transform under **Adjustments > Base > RAW Interpretation**. The selector belongs to the active variant and provides three policies:

- **Camera default — automatic fallback** first uses the explicit default set for that camera make and model, then a built-in profile qualified for automatic matching and, if neither exists, **Standard decoder camera matrix**;
- **Standard decoder camera matrix** forces the transform supplied by the RAW decoder for this variant only;
- choosing a named profile pins that exact artifact to the active variant. It does not alter the camera default.

The camera default is therefore a separate, model-wide policy. Setting or resetting it does not replace explicit choices already stored in variants. Of the bundled profiles, only the chart-measured Nikon D850 profile is currently qualified for automatic matching. Without an explicit camera default, other bodies fall back to the standard decoder matrix even when profiles not qualified for automatic matching are visible in the list. Bayer/X-Trans RAW files use LibRaw matrix metadata; supported X3F files use CAMF-derived metadata.

If an artifact pinned to a variant disappears or its contents change, ChromaLux does not silently substitute another profile: the variant safely falls back to the decoder matrix and reports that its profile is unavailable. A profile made for another camera can still be selected deliberately, with a color accuracy warning.

To measure a profile from a chart, see [Create a CC24 calibration profile](#).

Import a local profile

Import profile... accepts a compatible local DCP or SCSP. The file is qualified before installation and copied into ChromaLux's managed catalog; an accepted DCP becomes a managed SCSP object. Safe provenance retains the source name and hash, plus a provider when one can be identified, without storing the absolute local source path.

Every imported profile is marked **Experimental**, with usage rights **not verified**. ChromaLux infers neither a licence nor redistribution permission: the user must verify that the intended use is allowed. Import alone never applies a profile, never makes it eligible for automatic matching, and never promotes it to the camera default. Select it explicitly on a variant. After visual qualification, a separate and deliberate action can make it the explicit default for that camera model.

SCSP v3: bounded dual-illuminant DCP contract

SCSP v3 stores the exact validated DCP payload rather than flattening it to one matrix or LUT. ChromaLux can use its paired color and forward matrices, camera calibrations, analog balance, baseline exposure, hue/saturation maps, look table, and tone curve. At runtime it estimates the capture's correlated color temperature from the camera neutral and interpolates in reciprocal temperature (mired), with an explicit endpoint clamp rather than silent extrapolation.

The accepted contract is deliberately bounded: three-component camera RGB, supported illuminants with a deterministic CCT, and an SDR appearance payload. `ReductionMatrix` values are parsed and validated but are not used by the current three-channel solve. There is no independent tint interpolation axis. Malformed known tags, incomplete matrix pairs, singular transforms, incoherent tables, and unverifiable hashes are refused.

The **Import profile...** dialog and command-line tools accept this bounded contract. A successful import establishes structural and runtime-contract compatibility. Declared camera identity remains a separate diagnostic and may identify another body. Import does not establish color accuracy, a match to the provider's rendering engine, or rendering quality on a chart or independent RAW corpus.

Diagnose and compare from the command line

In a development or QA build tree, `cls_camera_profile_diagnostic` is read-only by default. It inspects a DCP or SCSP and keeps four verdicts separate:

structure, runtime compatibility, camera identity, and render qualification. The last one remains explicitly **not tested**:

```
./build-work/bin/cls_camera_profile_diagnostic --format json \  
--expected-make FUJIFILM --expected-model "GFX 100S" profile.dcp
```

The explicit `--import-local` option installs a qualified profile as **Experimental**, with unknown rights and no automatic selection. `--provider` records a provenance provider; `--catalog-root` isolates a QA catalog. Import requires both `--expected-make` and `--expected-model`.

```
./build-work/bin/cls_camera_profile_diagnostic --import-local \  
--provider "Local source" \  
--expected-make FUJIFILM --expected-model "GFX 100S" profile.dcp
```

For a develop A/B without changing preferences or sidecars, `cls_develop` accepts:

```
./build-work/bin/cls_develop --xmp --camera-profile auto -o auto.png  
image.raw  
  
./build-work/bin/cls_develop --xmp --camera-profile libraw -o libraw.png  
image.raw  
  
./build-work/bin/cls_develop --xmp --camera-profile profile.scsp -o  
profile.png image.raw
```

`auto` uses the camera resolver, `libraw` forces the standard matrix, and a `.scsp` path forces that artifact. An explicit SCSP is validated before rendering. These commands provide a reproducible comparison harness; they do not replace visual qualification on RAW files, highlights, and a chart.

SCSP v2: strict contract

An SCSP v2 profile must declare consistently:

- input primaries, white point, and transfer function;
- the post-white-balance input domain;
- output primaries, white point, transfer function, and domain;
- allowed numerical highlight headroom;
- calibration illuminant;
- provenance, generator, and source name;

- SHA-256 hashes of both the source and profile content.

ChromaLux rejects v2 when a mandatory field is missing, domains conflict, a hash does not match, content size is wrong, or a value exceeds declared headroom. Do not try to bypass rejection by renaming the file or editing its JSON.

The accepted v2 contract is intentionally narrow: white-balanced, linear camera-native input; linear working output with sRGB primaries and D65 white. Display conversion occurs later.

SCSP v1: ambiguous compatibility

Historical v1 profiles remain readable so existing developments do not break. They are classified **legacy ambiguous** because their metadata cannot fully establish primaries, white point, transfer function, headroom, and provenance.

ChromaLux never migrates v1 automatically to v2. A variant created with v1 continues using the historical contract. Profile Editor does not silently turn a v1 derivative into v2: it remains explicitly v1.

Warning — do not replace a production v1 with v2 merely because the file is newer. Compare independent image sets, retain both profiles, and create a new variant.

Arbitrarily converting a device ICC profile to SCSP does not prove that its ICC coordinates represent the expected linear camera RGB domain. The historical ICC-to-SCSP tool is disabled by default and is only for explicitly reproducing an old ambiguous v1 profile. Use Studio calibration for a new strict profile.

2. Choose the Viewer render space

Open **ChromaLux > Preferences... > Viewer > Color**, then choose **Viewer render space**:

- **sRGB** for general inspection and standard displays;
- **Adobe RGB (1998)** or **Display P3** for a calibrated display capable of reproducing that space.

This setting determines Viewer and cache values. Wider spaces enable 10-bit Viewer rendering when the hardware chain supports it. The Browser invalidates color-managed thumbnails when the space changes so it cannot reuse an old space's cache.

Viewer render space does not select the export recipe ICC profile. You can work in Display P3 and deliver sRGB JPEG, or work in sRGB and produce Adobe RGB TIFF, provided the final conversion is checked.

Output gamut mapping offers two contracts:

- **Legacy (default)** preserves the established terminal behavior;
- **Perceptual (OKLab v1) — Experimental** progressively compresses the most saturated colors toward the terminal RGB gamut.

Perceptual mapping applies only to v2/v3 chains whose output transform is truly terminal. A v1 image explicitly stays on Legacy. Mapping runs once before output encoding; export and Print consume that result without compressing it again. It replaces neither a recipe/printer ICC rendering intent nor soft proof simulation and gamut warning. The inner 80% of the radial OKLab gamut boundary is unchanged; the outer shoulder, including still-in-gamut saturated colors, is compressed. This experimental model still requires visual review.

SDR and EDR preview

Display range defaults to SDR. **EDR preview** is available only to a v3 variant on a compatible Apple display. It presents extended-linear Display P3 through an RGBA16Float surface, with paper white and creative peak bounded by the headroom advertised by the display. The main Viewer, Second Viewer, and Focus loupe use the conservative intersection of their display capabilities when they share one graph.

ChromaLux falls back to SDR when the surface, display headroom, luminance contract, pipeline version, or soft proof is incompatible. Moving between displays can therefore change the effective mode. EDR preview does not alter the export recipe: current exports and Print remain SDR, and ChromaLux does not yet encode HDR10/PQ or HLG files or delivery metadata.

3. Let ColorSync describe the display

On the same preference page, **Active display profile** shows the ColorSync profile attached to the display carrying the window; its tooltip includes the SHA-256 hash. ChromaLux updates this identity after a display or profile change.

The display profile belongs to presentation. It must not be embedded in the delivered file and does not replace either the camera profile or recipe ICC.

If ChromaLux reports the ICC profile unavailable:

1. move the window to the display being checked;
2. open **System Settings > Displays > Color** and inspect the ColorSync association;
3. recalibrate or attach a valid profile;
4. return to preferences and check name and hash again.

A displayed identity proves the profile was read, not that the monitor is currently calibrated. Record calibration date, instrument, target, and report outside ChromaLux.

4. Choose the delivery ICC profile

Each export recipe has its own **ICC** choice. ChromaLux starts from the actually rendered source space, converts pixels to the requested profile, and embeds it.

- **sRGB IEC61966-2.1** suits browsers, Web services, and common exchange.
- **Adobe RGB (1998)** suits some managed photographic workflows.
- **ProPhoto RGB** reserves a wider gamut but requires an ICC-aware destination and sufficient bit depth.

With no explicit ICC choice, export preserves the source render space rather than silently compressing to sRGB. Check the embedded profile before delivery. Prefer a compatible 16-bit format for a master when the downstream workflow accepts it; JPEG remains 8-bit.

Soft proofing is not baked into production export. It previews another destination.

5. Soft-proof a printed output

Soft proofing uses the real render-target ICC as source and the printer or press profile as simulated destination. Viewer and **Print** use one recipe contract:

- proof profile;
- perceptual or relative colorimetric intent;

- black-point compensation;
- paper-white simulation.

Simulation and **gamut warning** are separate calculations. Enable either, both, or neither. The warning marks out-of-proof-gamut areas in magenta; it changes neither simulated pixels nor the delivered file.

For a print decision:

1. calibrate the display and check its active ColorSync profile;
2. choose the exact printer–paper–ink profile;
3. enable proofing and choose the intent;
4. compare with and without paper-white simulation;
5. display gamut warning separately;
6. make a test print under controlled viewing light.

Warning — proofing does not establish physical print conformity. It depends on the profile, display and calibration, paper, ink, driver, and viewing light.

6. Use a neutral UI for color decisions

Under **ChromaLux > Preferences... > General > Appearance**, enable **Neutral color environment** to lock UI surfaces and accents to achromatic greys. Custom colors remain saved and return when the mode is disabled. Functional analysis overlays keep their colors.

A custom palette is applied only if text reaches at least 4.5 : 1 contrast against every UI surface. The ratio uses relative luminance after sRGB linearization. This helps readability; it calibrates neither the Viewer nor the display.

Check before delivery

Record:

- SCSP profile and version;
- Viewer render space;
- active ColorSync profile name and hash;
- ICC profile embedded in the file;

- proof profile and intent, if used;
- format, bit depth, and checking application.

See [Formats and compatibility](#), [Prepare a print](#), and [Initial troubleshooting](#).

Import, develop, and export

Documented version: ChromaLux 0.6.5

Product status: Stable

Documentation validation: PARTIAL

Evidence: import replayed on an earlier 0.6.5 package; development and export audited in code, translations, and tests

Blocker: new adaptive path, development, and delivery not replayed in the final package

English mirror: NOT REPLAYED

Outcome: a reproducible path from card to delivered file

This expands the [Quick start](#), with nondestructive handling and checks before any action that can affect source files.

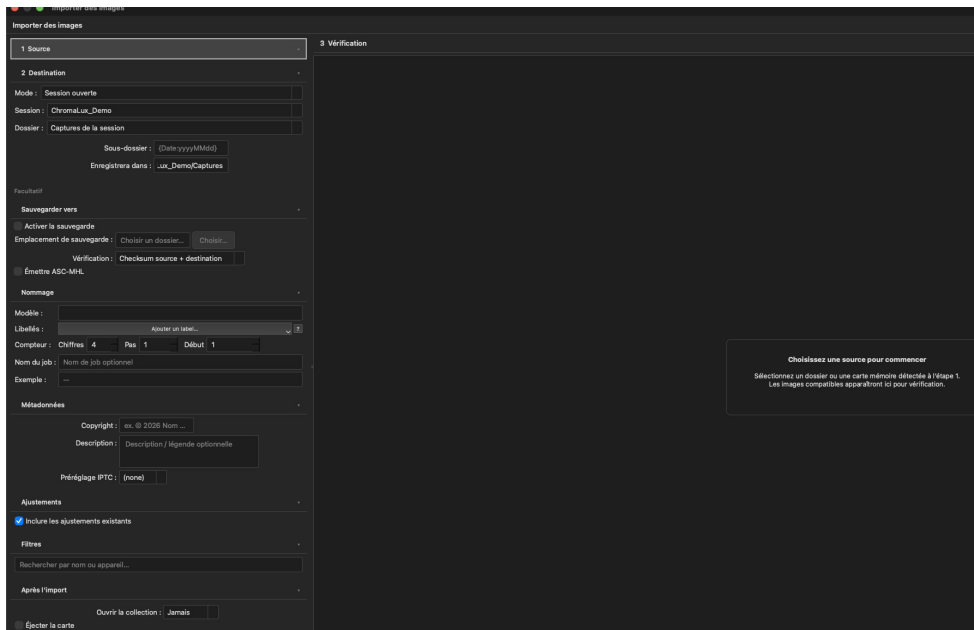
Prepare the session

Create a session with ⌘N or open one with ⌘O. A new session contains a `.clsession` catalogue and the **Captures**, **Selects**, **Output**, and **Trash** folders.

Photographer, client or project, campaign or series, job name, and production credits can feed metadata, naming, and captions when the relevant module offers them.

Import originals

Choose **File > Import Images...** (⌘I). The window has three visible stages: **Source**, **Destination**, then **Verification**.



The French source capture hides local volume identities. No English capture or replay is claimed.

1. Choose the source

Select a memory card, volume, or folder, or drag a folder from macOS Finder. Wait for inventory to finish and check the central grid before accepting the batch.

The review area distinguishes scan in progress, source with no importable image, filter with no result, and empty selection. **Import** remains disabled while scanning or with no candidate selected. The footer and accessible button description state the first blocking reason.

2. Choose the destination

For session work, choose a folder under **Captures**. Numbered folders created with / 00010 +10 12 can separate sequences or looks. Check the displayed final path; an unmounted or unwritable volume prevents import.

3. Set optional operations

Optional panels can:

- create and verify a backup copy;
- rename files using text and session labels;
- add common metadata.

Check the naming example, including counters, separators, and missing labels.

Use Tab / ⬆Tab from **Source** through **Destination**, advanced options, review grid, **Cancel**, then **Import**. Below 980 pixels wide, panels become one vertical column without changing that order.

Warning — Erase images after copying is destructive and off by default.

ChromaLux enables it only with **Source + destination checksum** and **Emit ASC-MHL**. Enter ERASE to arm it, then confirm again after import. If any batch file fails, the source remains intact. ChromaLux deletes only imported and verified originals; it does not format the card.

4. Run and check the import

Click **Import** and let progress finish. After a failure, do not format the card: record the message, check the destination, and retry only after finding the cause.

In **Library**, select the destination in ChromaLux Finder and compare the thumbnail count and names.

The 0.6.5 import scanner recognizes major Nikon, Canon, Sony, Fujifilm, Panasonic, Olympus/OM System, Pentax, Adobe DNG, Phase One, Hasselblad, GoPro, and Sigma/Foveon RAW families, as well as JPEG, HEIC, and HEIF. This archival list is broader than the set the Viewer can develop. Qualify unusual formats before production.

Make an initial selection

1. Navigate with ← and →.
2. Check sharpness with zoom or the Focus Loupe.
3. Apply ratings, color labels, or selection states.
4. Filter to isolate retained images.

The focused photograph is the source for adjustment and metadata copy.

Warning — selection synchronization is enabled by default. Reduce the selection to one photograph for a strictly individual adjustment.

Develop a RAW

Move from global rendering to local corrections.

1. Establish the base

In **Base**:

1. select the appropriate processing intent, camera profile, and demosaicing under **RAW Interpretation**;
2. neutralize or preserve the cast with White Balance;
3. set overall brightness with Exposure;
4. recover extremes with HDR without flattening the image;
5. inspect Detail at a relevant zoom;
6. adjust Primaries only when the color rendering requires it.

Use the histogram and clipping warnings, but also judge the image near its delivery size.

2. Build contrast and color

- Curves and Levels structure contrast.
- Tone Equalizer targets luminance ranges.
- Dehaze changes atmospheric contrast.
- Color Balance and Color Editor correct or direct the palette.
- Harmony organizes relationships among colors.

Make small changes and compare regularly with /. **BEFORE** shows the rendered RAW without user adjustments, not necessarily raw sensor bytes.

3. Localize only when needed

Use layers and masks only for corrections that must not affect the whole image. Fix a weak global base before masking around it.

4. Reuse a development

1. focus the source;
2. choose **Copy Adjustments** (⌘C);
3. select destinations;
4. choose **Paste Adjustments** (⌘V) and, if needed, limit adjustment groups from its menu.

Check at least one destination at 100%, especially across different exposures, ISO values, or crops.

Export with a recipe

A recipe combines format, size, location, naming, and delivery ICC profile.

1. Check each recipe to run.
2. Select a recipe to edit it.
3. Choose the destination folder.
4. Choose format, bit depth, and delivery **ICC** profile.
5. Optionally enter subfolder and job name.
6. Check the naming preview and final dimensions.
7. Click **Export**.

Several checked recipes create several outputs. + duplicates from the selected recipe; double-click or F2 renames it. A multiple selection exports every selected item with every checked recipe.

Resolution writes a metadata tag; only Scale changes pixel dimensions.

Recipe **ICC** is distinct from **Viewer render space**. ChromaLux converts from the actual render space to the requested profile and embeds it. With no replacement ICC, it preserves the rendered-space profile instead of silently compressing to sRGB.

JPEG is 8-bit. A 16-bit TIFF/PNG master remains high precision through its writer; another 8-bit recipe in the batch does not impose its quantization on the master.

Soft proofing is not baked into export. Empty queue offers **Export Current Image**; empty history offers **Open Destination Folder**. Progress exposes an updated accessible description.

Delivery check

- compare the output count with the expected count;
- check names, formats, dimensions, and embedded color profiles;
- open representative files, including the brightest and most saturated;
- keep the session, originals, and sidecars until delivery is accepted.

See [Initial troubleshooting](#). See also [Color management](#).

Organize and find session photographs

Documented version: ChromaLux 0.6.5

Product status: Stable, with search limits identified

Documentation validation: NOT REPLAYED

Evidence: audit of Browser model, selection, translations, and tests

Blocker: mouse, keyboard, and VoiceOver workflows not replayed in the final package

English mirror: NOT REPLAYED

Outcome: browse folders, build a selection, and find a file in the current folder

ChromaLux 0.6.5 organizes photographs primarily by folder. **Finder** chooses the folder; **Browser** shows images directly inside it.

Choose a working folder

Open **Library**, then use **Finder**:

- **Session** limits the tree to the open session;
- **All Files** browses other accessible folders and volumes;
- clicking a folder updates the Browser.

The Browser does not aggregate subfolders in 0.6.5. Select each subfolder to see its contents.

Role markers identify **Captures**, **Selects**, **Output**, and **Trash**. Right-click a folder to create a subfolder or assign one of these roles.

Move or copy between folders

Drag selected Browser thumbnails to ChromaLux Finder:

- a normal drag moves files;
- hold ⌘ while dropping to copy.

The target displays **MOVE** or **COPY** before the drop.

Warning — a move changes file locations on disk. Check the target and the displayed operation before releasing the mouse.

Move Folder to Trash... moves the folder and all contents to macOS Trash after confirmation. ChromaLux protects the session root and active role folders.

Build a Browser selection

Action	Gesture
Focus one photo and replace selection	Click
Add or remove one photo	⌘ + click
Select a continuous range	⇧ + click
Add a range without losing the existing selection	⇧⌘ + click
Select all displayed thumbnails	⌘A with Browser focused
Change focused photo	← or →

Clicking an already-selected photograph focuses it without dropping the others. Clicking empty Browser space clears the selection.

Rectangle or drag selection is unavailable in 0.6.5. Use modifier-clicks.

For assistive technologies, Browser separately announces selected count, focused photograph, whether it belongs to the selection, and the currently displayed photograph when different. This structure was code-audited; VoiceOver remains **NOT REPLAYED**.

Rate, pick, and label

Classification commands apply to all selected photographs. First focus the Browser.

Classification	Default shortcut
Rate one to five stars	1 through 5
Clear rating	0

Pick	P
Reject	X
Clear pick/reject	U
Grey, green, purple, blue, yellow, red, orange	⌘1 through ⌘7
Clear all color labels	⌘0

The context menu also provides pick/reject and color labels. A photograph can have several colors; mixed selections are identified. Ratings and statuses are stored in the session catalogue. Color labels also synchronize with macOS Finder tags; custom text tags are preserved.

Tip — reduce the selection to one photograph before an operation that must not affect the batch. The Browser bar shows the selected count.

Shortcuts can be changed under **View > Keyboard Shortcuts....**

Sort thumbnails

The 0.6.5 **Sort** menu supports:

- name;
- file date;
- rating;
- color label;
- ISO;
- file size;
- aperture;
- shutter speed;
- extension.

Reverse reverses the order. Names stabilize ties.

Lens, Focal Length, Width, Height, Exported, Sequence ID, and **Manual** remain visible but disabled; they are not available sort keys in 0.6.5.

Filter the current folder

The Browser-bar **Filter** field is the working search in 0.6.5:

1. choose a folder in Finder;
2. enter part of a name, extension, ISO, or date;
3. wait briefly for matching thumbnails;
4. clear the field to restore the folder.

Matching is case-insensitive.

Input	Expected result in current folder
portrait	file names containing “portrait”
nef	names or extensions containing “nef”
800	ISO containing “800”, and names containing “800”
2026-07-15	files whose modification date matches

This does not search subfolders, title, description, keywords, client, camera, or lens.

Search limits in 0.6.5

The Library **Filters** module still displays **Search**, **Rating ≥**, **Color**, and **Picks**. In the folder-based 0.6.5 Browser model, these controls do not reduce displayed thumbnails.

Show does work: choose **All**, **Originals Only**, or **Derivatives Only**.

Full-text title, description, keyword, and credit search is not delivered in this version. Build a verifiable batch by choosing the folder, using Browser **Filter**, sorting results, selecting with ⌘ or ⇧, and checking the count.

See [Metadata and naming](#) and [Import, develop, and export](#). See [Accessibility](#) for focus order and screen-reader limits.

Develop a RAW image

Documented version: ChromaLux 0.6.5

Product status: Stable, with experimental tools identified

Documentation validation: NOT REPLAYED

Evidence: audit of development pipeline, translations, and automated tests

Blocker: missing RAW reference corpus and final-package perceptual replay

English mirror: NOT REPLAYED

Outcome: a balanced, controlled, reusable development

Work from the starting rendering toward fine corrections so that a weak base is not hidden under accumulated tools.

Before you begin

Check that the RAW format is supported by the Viewer, initial decoding has finished, only one photograph is selected if you do not want batch synchronization, and histogram/clipping warnings are available.

See [Formats and compatibility](#) if a thumbnail appears but the image does not develop.

1. Choose the starting rendering

In **Adjustments > Base, RAW Interpretation** defines:

- **Processing intent:** full photographic workflow or Science without creative finishing;
- **Camera profile:** sensor-colour conversion into ChromaLux's working space;
- **RAW Demosaicing:** sensor mosaic to colour-image reconstruction.

Keep **Camera default — automatic fallback** and **HALO — recommended** without a specific requirement. **Effective transform** names the effective profile or shows **Standard decoder camera matrix**. Automatic fallback follows this order:

1. the explicit default set for that camera model;
2. a bundled profile qualified for automatic matching;
3. the standard matrix supplied by the RAW decoder.

Of the bundled profiles, only the chart-measured Nikon D850 profile is currently qualified for step 2. Other bodies therefore use the decoder matrix unless an explicit camera default has been set. This is LibRaw metadata for Bayer/X-Trans RAW files and CAMF-derived metadata for supported X3F files.

Choosing a named profile or **Standard decoder camera matrix** is stored on the active variant, not as the model default. The camera default is set and reset separately; changing it does not replace explicit choices in existing variants. You can deliberately select another camera's profile, but ChromaLux marks it as an expert choice that may produce inaccurate colors.

Compare with decoder matrix temporarily presents the same variant through that standard matrix. This A/B belongs only to the focused Viewer: it is written to neither the variant, sidecar, nor camera default, and is dropped when the image, variant, or focus changes. To retain this fallback, explicitly select **Standard decoder camera matrix**.

Import profile... adds a compatible DCP or SCSP to the local catalog as **Experimental**, with unverified rights and recorded provenance. Import never applies the profile or adds it to the built-ins qualified for automatic matching; select it explicitly on the variant being evaluated. Import compatibility is not render qualification: compare multiple RAW files, skin, saturated colors, and highlights.

Defringe belongs to Bayer demosaicing. Do not change the camera profile just to fix a dark or cold photograph; use Exposure and White Balance.

2. Balance neutral colors

In **White Balance**:

- negative **Temperature** cools; positive values warm;
- negative **Tint** corrects toward green; positive toward magenta;
- **Auto WB** progressively neutralizes the measured cast, mainly in midtones;
- **Cast** moves measured colors toward neutrality.

Use the picker only on a genuinely neutral surface under the light being corrected. A white object under colored light or a painted wall is not a reliable neutral.

3. Set global brightness and contrast

Setting	Effect	Watch for
Exposure	Moves global brightness in photographic values; +1 EV doubles scene luminance.	Strong increases can clip highlights.
Balance	Couples density and color intensity; positive is darker/ richer, negative lighter/washed.	It is not a second exposure control.
Contrast	Separates or brings tones together around middle grey.	Excess closes shadows and hardens skin.
Brightness	Mainly lifts midtones while maintaining black and white.	Prefer it when extremes are already placed.
Saturation	Changes all colors without changing luminance.	Vivid colors can leave the output gamut.
Vibrance	First strengthens less-saturated colors and protects neutrals.	It does not replace targeted correction.

Set **Exposure**, then **Brightness** and **Contrast**. Use **Balance** only for an intentional density/color relationship.

4. Restore dynamic range

In **Extended Dynamic Range**:

- lower **Highlights** to recover bright detail; linked mode also engages clipped-channel reconstruction;
- raise **Shadows** to open dark tones while holding black;
- raise **White** for sparkle or lower it to soften the upper range;
- lower **Black** for deeper blacks or raise it for a lifted matte black.

Raised shadows can reveal noise. Fully clipped highlights cannot contain the same information as merely bright areas.

Advanced recovery

Use **Advanced Recovery** only when normal highlight correction is insufficient:

- **Recovery** combines channel reconstruction, density, and adaptive hue;
- **Channel** reconstructs a clipped channel from remaining channels/details;
- **Spatial** borrows nearby color for fully clipped areas;
- **Density** advances highlight compression;
- **Whitening** reduces color toward white;
- **Hue** lightly colors reconstructed neutral highlights;
- **Shadow Detail** restores texture near the noise floor.

High values can invent color, spread a hue, or reveal noise. Check multiple zoom levels.

5. Set detail at 100%

Judge **Detail** at 100%, then at delivery size.

Sharpening

- **Amount** increases edge contrast; excess produces halos and graininess.
- **Radius** controls edge width.
- **Chroma Boost** has no effect with the Richardson–Lucy sharpening engine used by default in 0.6.5.

Presence

Texture, **Clarity**, and **Structure** affect progressively larger detail scales. Start low on portraits and watch skin and light transitions.

Noise reduction

- **Luminance** reduces brightness variation;
- **Color** reduces color blotches while protecting colored edges;
- **Shadow Detail** restores some protected texture.

The noise-profile label states whether ChromaLux uses a camera measurement or a generic ISO-based profile.

6. Refine tones

- **SCS Curves** shapes luminance, saturation, or RGB channels.
- **SCS Levels** sets black/white points and midtone balance.
- **Tone Equalizer** targets eight tonal ranges.
- **Dehaze** restores atmospheric contrast; **Depth** locates it and **Warm-tone Protection** protects skin and sunsets.
- **Fringes** targets purple or green edges by threshold, hue range, and fade.
- **Vignette** lightens or darkens edges and controls center, softness, shape, and position.

Use the fewest overlapping tonal tools possible.

7. Correct or direct color

Color Balance separates shadows, midtones, and highlights. **Color Editor** targets a color family. **Harmony** guides dominant colors while allowing skin and neutral protection.

Primaries changes the global response of red, green, and blue primaries. Use it to construct an overall rendering, not to fix one isolated garment.

8. Localize a correction

Use **Layers & Masks** for a region-only adjustment. Version 0.6.5 provides Brush, Linear and Radial Gradient, SCS Local Point, Parametric SCS, and Chromatic masks. Brush masks can combine Apple Vision and SAM selections with SCS edge refinement.

See [Layers and masks](#).

9. Check the result

- J toggles output clipping.
- ⬆J toggles pre-recovery sensor clipping by channel.

- F toggles the SCS false-color map.
- / toggles the rendered RAW without user adjustments.
- \ bypasses Science-analysis overlays.
- Histogram shows tonal distribution.
- Vectorscope shows hue and saturation distribution.

10. Reuse and protect the development

- undo with ⌘Z, redo with ⇧⌘Z;
- bypass a module with its power button;
- reset a module from its header;
- reset the complete focused photograph with ⌘R;
- copy with ⇧⌘C, then paste to selected destinations with ⇧⌘V.

Check at least one destination after paste, especially when exposure, ISO, or light differs.

See [Color management](#), [Keyboard shortcuts](#), and [Initial troubleshooting](#).

Work with layers and masks

Documented version: ChromaLux 0.6.5

Product status: Stable for layers and manual masks; AI selection availability depends on macOS and installed models

Documentation validation: PARTIAL

Evidence: code and translation audit; 0.6.5 captures checked in an earlier PDF generation

Blocker: incomplete keyboard workflow, AI selections, and final-package visual replay

English mirror: NOT REPLAYED

Outcome: apply a development to a precise area without changing the rest of the image

Layers localize adjustments. A white mask area is processed, black is protected, and grey receives a partial correction. Set global exposure, white balance, and contrast before creating a layer.

Before you begin

1. Select the photograph.
2. Open **Adjustments**.
3. Choose **Layers & Masks**.
4. Keep **Image Layer (Background)** selected while editing global development.

Adjustment layers stack above the background and can be enabled separately.

1. Create and organize a layer

The + menu offers:

- **Add New Empty Adjustment Layer;**
- **Add New Filled Adjustment Layer;**

- layers with **Brush**, **SCS Local Point**, **Radial Gradient**, or **Linear Gradient**.

The new layer becomes active. Compatible modules then target it: Exposure, White Balance, Color Editor, Color Balance, Primaries, Fisher Curve, Black & White, and Extended Dynamic Range / Tone Map.

For global work, reselect **Image Layer (Background)**. On a selected layer, toggle enabled state, set **Opacity** from 0 to 100, choose **Normal**, **Multiply**, **Screen**, **Overlay**, **Soft Light**, **Difference**, **Add**, or **Subtract**, and use – to delete.

The stack and its controls have accessible names. Arrow keys change the selected layer and Tab moves into controls. + and – targets are 28 × 28 pixels.

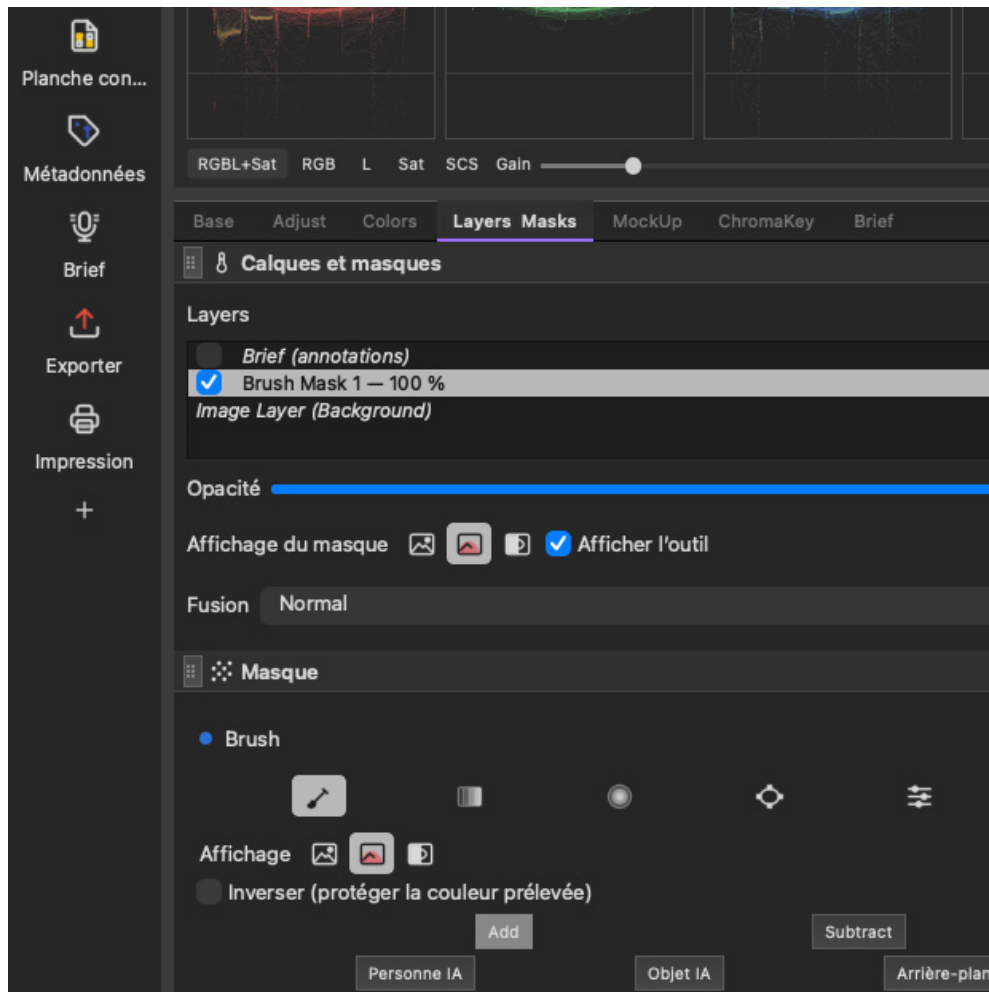
0.6.5 limit — **empty** and **filled** adjustment-layer commands still create the same initial graph with no correction. Use one of the four mask-creation commands for an immediately localized area.

Clone and **Heal** are visible as future features but disabled.

2. Choose a mask type

The image stack accepts at most eight mask entries. The selector order is:

1. **Brush**;
2. **Linear Gradient**;
3. **Radial Gradient**;
4. **SCS Local Point**;
5. **Parametric SCS**;
6. **Chromatic**.



The source guide supplies this French 0.6.5 capture; no English capture is claimed.

Brush

Paints or erases freely; use it for manual correction, AI cleanup, and irregular shapes.

Linear Gradient

Transitions between zero-weight and maximum-weight lines. Move Viewer handles to orient and widen it.

Radial Gradient

Creates an elliptical transition around an editable center, with separate horizontal/vertical radii, feather, and rotation.

SCS Local Point

Combines an ellipse or rectangle with perceptual SCS selection. Center, radii, feather, and rotation are independent. Diagnostic views isolate final mask, SCS distance, spatial shape, hue, Fisher luminance, chroma, or legal range.

Parametric SCS

Builds from perceptual hue, Fisher luminance, and chroma ranges. Use feathering to avoid abrupt transitions.

Chromatic

Selects colors near up to four sampled hues. Sample size can be 1×1, 3×3, 5×5, or 9×9; larger samples are often steadier on noisy RAW data.

3. Inspect the mask

Mask display	Use
Image	Normal photograph, no mask visualization.
Overlay	Mask over the image for coverage and edge checking.
Black & White	Matte only: white processed, black protected, grey partial.

These are transient Viewer presentations. They change neither development nor sidecar, thumbnails, or production exports.

- M shows/hides the active mask editing tool.
- ↑M cycles **Image**, **Overlay**, **Black & White**.

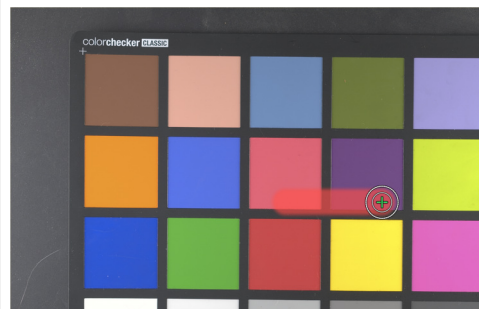
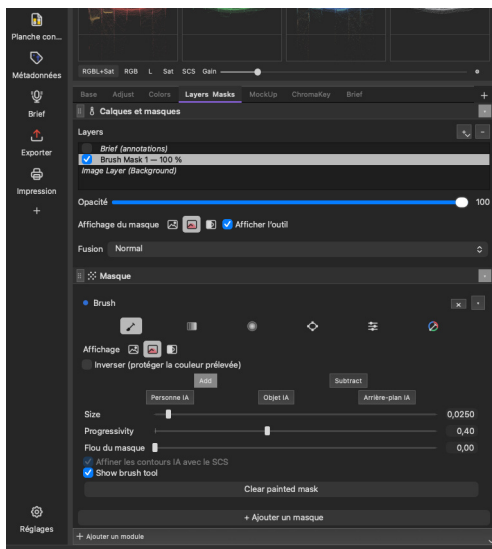
Show Tool separately controls handles, guides, or active-mask cursor. Each entry exposes type, enabled state, and collapsed/expanded state. Tab / $\uparrow$ Tab move among controls and Space activates a focused control. Painting and Viewer-handle movement still require a pointer.

4. Paint and correct a mask

Create **Brush**, enable **Show Tool**, then choose **Add** or **Subtract**. Each stroke retains the size and feather used when drawn.

Gesture	Effect
Hold ⌘ while painting	Temporarily subtract.
Hold $\text{⌘}\text{⌘}$, drag left/right	Decrease/increase size.
Hold $\text{⌘}\text{⌘}$, drag down/up	Decrease/increase hardness.
$\text{⌘}Z$ / $\uparrow\text{⌘}Z$	Undo / redo.

Size is relative to the image long side; **Feather** softens new strokes; **Mask Blur** applies nondestructive Gaussian blur to the complete mask. **Mask Blur** ranges 0–100 and represents sigma in full-resolution pixels. Check the black-and-white matte at high values.



French source capture. Add/Subtract remain available after Apple Vision or SAM selection.

Clear Painted Mask removes the AI base plus all add/subtract strokes. Use ⌘Z to undo only the latest action.

5. Select with Apple Vision

Brush offers:

- **AI Person**, selecting the person instance under a click/stroke;
- **AI Background**, selecting the inverse of detected foreground subjects.

They require Apple Vision on macOS 14 or later. ChromaLux does not silently replace them with SAM.

For a person, choose **AI Person**, click/stroke the subject, wait, inspect **Overlay** or **Black & White**, then correct with **Add / Subtract**. Successive Vision selections are additive and can combine with SAM.

⌘ negative prompts are unavailable for these Vision tools; subtract after selection.

6. Select an object with SAM 2.1

AI Object uses SAM 2.1 Small through Core ML:

1. start with a positive click/stroke on the object;
2. add positive clicks for missing parts;
3. hold ⌘ while clicking to exclude a region;
4. finish with manual Add/Subtract.

SAM accepts at most fourteen prompts. Running AI Object again replaces the previous SAM contribution while preserving Vision and manual contributions.

Install SAM models

The models are not bundled. Put these folders under:

~/Library/Application Support/ChromaLux/Models/apple-coreml-sam2.1-small

- SAM2_1SmallImageEncoderFLOAT16.mlpackage;
- SAM2_1SmallPromptEncoderFLOAT16.mlpackage;

- SAM2_1SmallMaskDecoderFL0AT16.mlpackage.

Advanced installations can set CLS_SAM21_MODEL_PATH to another parent folder. If any package is missing, **AI Object** is unavailable.

SAM input's long side is capped at 2048 pixels, then projected to the full mask. Check small details and fine edges.

7. Refine AI edges with SCS

Refine AI Edges with SCS constrains refinement to a narrow band around the Vision/SAM proposal. SCS compares local color and Fisher luminance.

It does not replace segmentation: Vision/SAM defines the region, SCS adjusts a local boundary, and Add/Subtract resolves ambiguous, transparent, or very fine areas. Compare on/off when subject and background colors are close.

8. Recommended method

1. finish global work on **Image Layer (Background)**;
2. choose the closest mask shape;
3. use **Overlay** for the overall area;
4. briefly use **Black & White** for holes, noise, and spill;
5. apply an intentionally visible adjustment;
6. correct the mask;
7. reduce adjustment and layer opacity to final values;
8. return to **Image** and hide the tool with M.

See [Develop a RAW](#), [Accessibility](#), and [Keyboard shortcuts](#).

Correct lens distortion and perspective

Documented version: ChromaLux 0.6.5

Product status: geometric correction using an exact lens profile or manual settings; nondestructive perspective correction

Documentation validation: PARTIAL

Evidence: audit of code, translations, automated tests, and the 0.6.5 capture

Blocker: final visual replay and keyboard movement of Perspective guides remain incomplete

English mirror: NOT REPLAYED

Outcome: straightened lines, controlled perspective, and final crop without changing the source file

Lens distortion and perspective have different causes. Apply **Lens Correction**, then **Perspective**, then **Crop**.

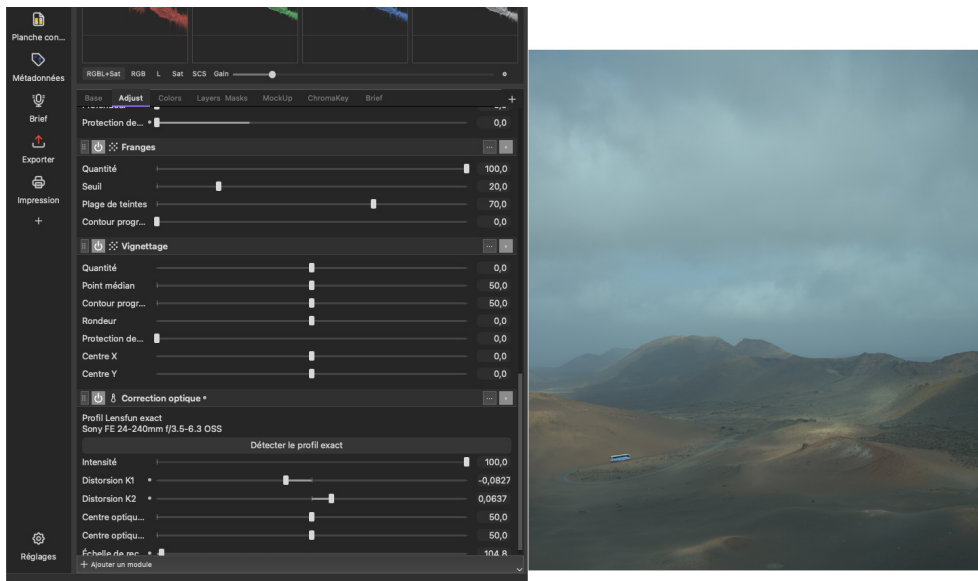
Correct lens distortion

Open **Adjustments > Adjustments > Lens Correction**. ChromaLux first looks for an embedded DNG correction, then for one exact, unambiguous match in the bundled Lensfun database.

The summary reports:

- **Embedded DNG correction** for compatible file data;
- **Exact Lensfun profile** for an exact camera/lens/capture match;
- **Manual correction** with no resolved profile;
- **modified** after manual adjustment of a resolved profile.

Use **Detect Exact Profile** to rerun matching. ChromaLux does not choose a “near” profile. Missing, ambiguous, or not-exactly-representable DNG correction is refused rather than approximated or replaced by a second Lensfun correction.



Refine the result

Setting	Use
Intensity	Blend between original and resolved geometry.
Distortion K1	Refine the main radial term.
Distortion K2	Refine the higher-order radial term.
Optical Center X/Y	Move correction center.
Auto-crop Scale	Enlarge to remove empty corners; at 100%, retain the whole corrected field.

Check straight lines near edges. Set K1 first, K2 only for changing residual curvature, and optical center only for asymmetry. Higher auto-crop removes corners but reduces the visible field.

Refusal and reproducibility

Malformed embedded data, per-channel variation, unsupported tangential terms, or several exact Lensfun candidates are refused. The Viewer stays uncorrected until a manual correction is set.

Once resolved, a numerical coefficient/provenance snapshot is stored in the variant sidecar, so later Lensfun changes do not alter it. Unsafe or invalid snapshots are ignored and correction remains off.

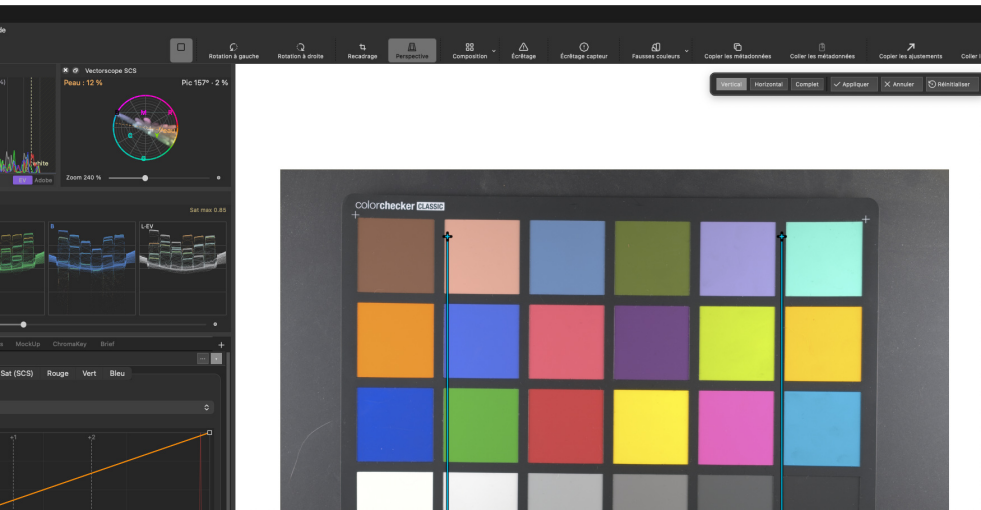
0.6.5 limit — this module corrects geometric distortion only. It does not apply Lensfun vignetting or lateral chromatic aberration. DNG per-channel corrections requiring those are refused, not approximated.

Straighten perspective

Perspective is distinct from **Crop**:

Mode	Guides	Points
Vertical	two vertical lines	four endpoints
Horizontal	two horizontal lines	four endpoints
Complete	two vertical and two horizontal lines	eight endpoints

Place guides along scene lines that should be straight. Complete mode requires coherent, nondegenerate geometry. Switching mode preserves all eight points while showing only active guides.



- click **Apply** or press Return;
- click **Cancel** or press Escape;
- click **Reset** to clear saved perspective correction.

Keyboard navigation limit

Use Tab / ⬆Tab in the palette. Return applies; Escape cancels.

Keyboard status — PARTIAL. Guide endpoints cannot yet be fully selected and moved by keyboard. Accurate placement requires a pointer. Apply/cancel shortcuts do not constitute a complete keyboard workflow.

Coincident guides, incoherent intersections, or non-invertible transformations are refused; the previous saved geometry remains intact. Guide positions and transform are stored deterministically in the variant sidecar.

Preview while calculating

Entering or applying Perspective can recalculate high-definition tiles. ChromaLux retains a full-image proxy while preparing tiles offscreen, then replaces it. Crop uses the same transition. Wait for high-definition rendering before judging detail and edges pixel-for-pixel.

Final check

1. check lines near center and all edges;
2. check for empty corners;
3. inspect at 100% after high-definition tiles return;
4. crop only then;
5. temporarily bypass **Lens Correction** to compare without resetting it.

See [Accessibility](#).

Metadata and naming templates

Documented version: ChromaLux 0.6.5

Product status: Stable

Documentation validation: NOT REPLAYED

Evidence: audit of code, translations, and XMP formats

Blocker: writing, copying, and naming templates not replayed in the final package

English mirror: NOT REPLAYED

Outcome: identify a production, describe photographs, and build names or captions from reusable labels

ChromaLux separates production-wide session information from per-photograph metadata.

Enter session information

Open **Metadata > Session Information**:

1. enter **Photographer**, **Client / Project**, **Campaign / Series**, and **Job Name**;
2. use **Add Field...** for assistant, Digital Tech, makeup, hair, stylist, stylist assistant, art director, creative director, or set decoration;
3. enter each person or studio;
4. click **Save Session Information**;
5. check the confirmation.

Removing a credit with × removes it from session information when saved.

Warning — changes do not retroactively update previously imported or captured photographs, or an already-loaded contact sheet. Set them before the relevant import, capture, or sheet.

Read photograph metadata

Photo Metadata groups:

- descriptive and XMP data;
- production credits;
- capture/EXIF data;
- file and associated XMP location.

With a multiple selection, the module shows only the focused photograph and names it at the top. Capture fields, path, rating, colors, and pick status are read-only here. Change classification in the Browser.

Edit the focused photograph

1. focus the photograph in the Browser;
2. click **Edit** in **Photo Metadata**;
3. edit title, description, or other fields;
4. separate keywords with commas;
5. click **Save**;
6. wait for reload and check displayed values.

Editable fields are title, description, keywords, photographer, client or project, campaign or series, job name, copyright, and production credits. Saving updates the session catalogue and creates or updates the adjacent XMP: DSC_0123.NEF.xmp.

Whitespace and empty keywords are removed. Case-only keyword duplicates are merged.

Warning — Edit targets the focused photograph even when several are selected. Use **Keywords** for a shared keyword, or **Copy Metadata** / **Paste Metadata** for the descriptive set.

If existing XMP is unreadable or malformed, ChromaLux refuses the write and leaves it intact. Back up or repair it before retrying.

Add or remove keywords from a batch

The **Keywords** module targets the complete Browser selection:

1. select photographs;
2. enter a keyword;

3. click **Add** for every selected photograph;
4. enter/select it and click **Remove** to remove it from the batch.

The visible list is the intersection of keywords common to every selected photograph, not their union. Catalogue and XMP are updated for each item.

Warning — Remove deletes the keyword from every selected photograph. Check the selection count first.

Copy metadata between photographs

1. focus the source;
2. click **Copy Metadata**;
3. select destinations;
4. click **Paste Metadata**;
5. check the updated-count message.

Paste replaces title, description, keywords, photographer, copyright, production identity, and credits. It does not change EXIF, rating, color labels, or pick status. The metadata clipboard is cleared when sessions change.

Metadata added at import

For an import into a session, ChromaLux adds non-empty session identity and credits to new XMP files. The import **Metadata** section can also add description, copyright, and an IPTC preset containing creator and keywords. Session **Photographer** takes precedence over preset creator.

0.6.5 limit — this describes **Import Images**. Card-insertion automatic import copies session identity but does not yet connect its preset's IPTC, copyright, and description.

Metadata added during tethered capture

Every finalized tethered file receives non-empty active-session information in XMP. The rule covers RAW+JPEG pairs. Session changes apply to subsequent captures and do not rewrite earlier ones.

Build a naming template

Import **Naming** and tether **Next Capture Naming** share one label editor:

1. place the cursor in **Template**;
2. type fixed text, for example CLIENT_LOOK_;
3. use **Add Label...**;
4. drag labels to reorder;
5. set counter **Digits**, **Step**, and **Start**;
6. optionally enter **Job Name**;
7. always inspect **Example**.

With start 10, step 10, and five digits:

```
{ClientProject}_{cnt:05}
```

```
MAISON_AZUR_00010
```

```
MAISON_AZUR_00020
```

```
MAISON_AZUR_00030
```

Forbidden file-name characters are cleaned. An unknown label can remain literal; look for braces in **Example**.

Available naming labels

File and sequence

Label	Value
{OrigName}	original name without extension
{OrigExt}	original extension, including the dot
{OrigFolder}	original folder when supplied by context
{Job}	Job Name
{cnt:04}	four-digit counter in this example
{Date:yyyyMMdd}	date in the stated format

{SessionName}	session or destination name supplied by the module
---------------	--

Camera

Label	Value
{CamMake}	camera manufacturer
{CamModel}	camera model
{ISO}	ISO sensitivity
{Lens}	lens
{Focal}	rounded focal length in millimetres

Session and production

{Photographer}	{ClientProject}	{CampaignSeries}
{BudgetName}	{Assistant}	{DigitalTech}
{Makeup}	{Hair}	{Stylist}
{AssistantStylist}	{ArtDirector}	{CreativeDirector}
{Deco}		

Known empty labels disappear. Design separators so they do not double, and always inspect **Example**.

Operation-specific limits

Label group	Import 0.6.5	Tethered capture 0.6.5
{OrigName}, {Job}, {cnt:...}	available	available
session/production labels	available	available
{OrigExt}	avoid: import already	available

	appends extension	
{OrigFolder}	not supplied	available
{Date:...}	current date, not source EXIF	capture time
camera, ISO, lens, focal length	not supplied	available if camera service supplies them

The editor exposes a common vocabulary, but contexts do not supply every value. EXIF-to-import naming is not connected in 0.6.5; do not use camera labels to rename a production import.

Reuse labels in a contact sheet

Enter session information before creating the sheet. In **Contact Sheet > Logo & Caption**, place the cursor in **Caption**, click +, select a Session or Production Team label, combine with text, reorder, and check preview.

Contact-sheet captions accept only session and production labels, not file, counter, date, or camera labels.

See [Import, develop, and export](#) and the [local AI metadata appendix](#).

Build a look with Film Simulator

Documented version: ChromaLux 0.6.5

Product status: Stable

Documentation validation: NOT REPLAYED

Evidence: audit of code, translations, and film data

Blocker: perceptual replay on reference images in the final package is missing

English mirror: NOT REPLAYED

Outcome: a coherent, adjustable, reusable film rendering without changing the original

Film Simulator combines tonal/color response with black and white, halation, darkroom print, grain, and vignette.

Before you begin

Finish neutral work in **Adjustments**: RAW Interpretation and camera profile, White Balance, Exposure/highlight recovery, and required noise or lens corrections. Crop before placing a vignette precisely.

Warning — selection synchronization is enabled by default. Reduce the selection to one photograph if the look must not propagate.

Recommended order

1. Choose the stock under **Film Curve**.
2. Set intensity, brightness, and contrast.
3. Refine **Black & White** if monochrome.
4. Set **Halation** on highlights.
5. Add **Darkroom Print** when wanted.
6. Inspect **Film Grain** at 100%.
7. Finish with **Vignette** after crop.

Choosing a stock in **Film Curve** also synchronizes associated **Black & White**, **Film Grain**, and **Halation** settings. Choose it first because it can replace manual settings in those modules.

1. Choose and scale the film curve

Standard mode

Choose a **Film Stock**, then adjust:

Setting	Effect
Intensity	Scales tonal/color character.
Contrast	Strengthens or softens light/dark separation.
Brightness	Changes the perceived lightness without replacing technical Exposure.

Some stocks offer **Development: Legacy (Y-luminance cascade)**, **Pull-1**, **Normal**, or **Push-1**. It appears only when the stock has the required data.

Expert mode

Expert exposes density shape:

- **Toe Density** for shadow entry;
- **Shoulder Density** for highlight compression;
- **Gamma** for main slope/mid-contrast;
- **D max** for maximum density;
- **Grain (µm)** and **Polydisp.** for the curve's model.

With a stock selected, its color response remains while Expert supplies the tonal shape. **Custom** builds the shape manually without a stock-specific gamut.

0.6.5 UI limit — an Expert tooltip may say “without preset” even though the selected stock retains color response. Another presents scan calibration as future although the catalogue already includes scan-derived responses. This page describes the shipped engine.

Warning — choosing **Custom** does not zero sliders. It retains displayed values.

The 0.6.5 catalogue mixes scan-calibrated stocks with initial models based on technical data or estimates. A name is not an exhaustive measurement of one physical film, lab, and scanner combination.

2. Build black and white

Enable **Black & White** from its header.

Color sensitivity

Red, Yellow, Green, Cyan, Blue, and **Magenta** determine how original colors become greys. Positive values lighten a family; negative values darken it. Avoid artificial transitions.

Split toning

Set highlight and shadow hue separately, then raise saturation slightly.

A Film Curve stock can enable this module and load sensitivity. Some monochrome stocks already contain a black-and-white color response; ChromaLux avoids converting twice.

The module has no ... menu in 0.6.5. Use global adjustment copy to transfer it.

3. Set halation

Setting	Effect
Threshold	Restricts the effect to brighter areas.
Diffusion px	Apparent halo radius.
Amount	Visibility.
Tint R, Tint G, Tint B	Halo color.

The module's **Film Stock** menu loads only a halation proposal. Start with low **Amount** and inspect lamps, windows, and specular highlights. A low threshold spreads haze into midtones.

4. Simulate a darkroom print

Darkroom Print is an in-image creative finish, not page layout or printer control. Choose **Glossy baryta**, **Neutral RC**, **Matte**, **Selenium**, **Warm sepia**, or **Soft matte**.

Presets initialize **Finish**, **Diffusion**, **Tonal Compression**, **Halo**, **Paper Warmth**, **Toning**, and **Paper Texture**. After manual change, **Custom** appears when values no longer match a shipped paper.

Warning — choosing **Custom** neither resets nor enables the module.

Use the separate [Print workflow](#) for physical output.

5. Set film grain

The module's stock menu loads only a grain proposal; it does not change Film Curve.

Setting	Effect
Grain (µm)	Simulated physical grain size.
Polydisp.	Distribution of grain sizes.
Intensity	Visibility.
Seed	Random arrangement without changing general style.
PSF Blur µm	Optical-like softening.
Chroma Strength	Colored grain contribution.
Crosstalk	Grain response mixing among channels.

Inspect at 100%, then at delivery size; resizing changes perception.

6. Finish with vignette

Film Simulator **Vignette** is creative and distinct from technical lens vignetting correction.

- negative **Amount** darkens edges, positive lightens;
- **Midpoint**, **Feather**, and **Roundness** shape it;
- **Highlight Protection** preserves bright edge areas;

- **Center X/Y** move its center.

Set it after crop.

Check the look

Check at fit-to-window, 100%, and delivery size. Use / to compare with the rendered RAW without user adjustments.

Reuse on a series

Focus a reference, copy adjustments, and retain **Black & White** plus Film Simulator's **Film Curve**, **Film Grain**, **Halation**, and **Darkroom Print** groups. Include **Vignette** from its lens/vignetting group when part of the look. Select destinations, then paste.

The ... menus can separately save or copy Film Curve, Film Grain, Halation, Darkroom Print, and Vignette.

Warning — check the selection before pasting; the command can target every selected photograph.

Common problems

My grain or halation settings changed

Select the film stock before fine-tuning its associated effects.

Grain is invisible or too strong in export

Inspect at 100%, then at final delivery size.

Halation colors too many elements

Raise **Threshold**, lower **Amount**, then set **Diffusion px**.

Moving modules does not change the result

This is expected. The processing graph is fixed; panel movement changes layout only.

For more information

See [Develop a RAW](#) and [Selection and nondestructive editing](#).

Work in tethered capture

Documented version: ChromaLux 0.6.5

Product status: Stable — Studio feature; compatibility depends on camera

Documentation validation: PARTIAL

Evidence: code, translation, and transaction-test audit; earlier hardware observations

Blocker: camera, firmware, and provider matrix not replayed in the final package

English mirror: NOT REPLAYED

Outcome: trigger, download, and name a photograph from a USB-connected camera

Compatibility depends on model, firmware, USB mode, and provider. ChromaLux 0.6.5 contains Nikon, Canon, Fujifilm, and explicit gPhoto2 fallback paths. Presence is not certification of every camera.

Earlier tests observed Nikon D850 and Fujifilm X-T4 with libgphoto2 2.5.34 in signed staging. These are historical observations, not qualification of current vendor services or the final package.

Before connecting

1. Open a ChromaLux session.
2. Check capture folder and free space.
3. Connect directly by USB where possible.
4. Use sufficient battery or suitable power.
5. Close Photos, Image Capture, and vendor applications that can own USB.
6. Set a camera USB mode compatible with tethering.

Make a no-stakes capture first, then check RAW, name, folder, and metadata.

Prepare folder and naming

In **Tethered Capture**:

1. keep **Connect Automatically** enabled for automatic detection/opening;
2. choose **Capture Folder** for subsequent photographs;
3. choose or enter a naming template;
4. enter the free-form shoot name when used;
5. check **Next** and the counter;
6. use **Reset** only when the next number must restart.

Session and production labels are available. Empty session values produce empty name components.

Choose the provider

- **Automatic — recommended** detects Nikon, Canon, or Fujifilm USB vendor and starts its service.
- **Nikon (Z / D850), Canon (EOS R), or Fujifilm (X / GFX)** forces that service.
- **Generic gPhoto2 — manual fallback** uses libgphoto2.
- **Mock** is synthetic testing, not production.

Automatic mode never silently replaces a vendor provider with gPhoto2. Disconnect the current service before explicitly trying the fallback.

Connect and open the camera

If automatic connection is off, click **Connect**, wait for **Ready**, use **Refresh** if needed, choose the camera, and watch status until its session is open. Capture can implicitly open the camera on the first shutter action. Buttons remain disabled when the provider does not report the capability.

Use Live View

Start Live View is available only after open and when supported. It opens a dedicated window; **Show in Viewer** mirrors the stream. Main Viewer, this window, and **Raw Mockup** share one camera stream rather than opening three connections.

If unavailable, check open state, disconnect, explicitly try **Generic gPhoto2**, reconnect, and inspect capabilities. Missing Live View does not necessarily prevent

capture. Only one application should own USB.

Trigger and check a capture

1. enable **Select new captures automatically** if desired;
2. trigger from ChromaLux or the camera when supported;
3. wait for every file to download;
4. inspect **Captures** and optionally **Show in Finder**;
5. check the RAW in the Viewer before the next shot.

A capture can produce RAW and JPEG. ChromaLux waits for completion and expected files, then verifies/deposits the transaction as one set to prevent partial or duplicate publication.

All files are imported. RAW is primary in a RAW+JPEG transaction; automatic selection focuses that first file without hiding its JPEG companion.

Change camera settings

Camera Settings shows only properties the provider can list and change. A refused or unconfirmed value is restored in the UI. Change sensitive modes, autofocus, drive, or format on the camera when no clear UI equivalent exists.

Disconnect cleanly

1. stop Live View;
2. wait for downloads;
3. click **Disconnect**;
4. power off before unplugging.

Earlier hardware observations

Nikon D850

Phase 7 observed shutter and Live View through gPhoto2 on firmware 1.31. If macOS reclaims USB, close Photos and Image Capture, power-cycle, then reconnect.

Fujifilm X-T4

Phase 7 used firmware 2.12, **AF-S**, and single-shot. Manual focus with timer was refused in those tests. For a persistent busy state, disconnect/reconnect after closing other photo applications.

Export diagnostics

Export Tethering Diagnostic... writes JSON describing service, camera, capabilities, and recovery information, with no photograph. Attach camera model, firmware, macOS version, and reproduction steps. A gPhoto2 “compatible” label is not ChromaLux certification.

Create a CC24 calibration profile

Documented version: ChromaLux 0.6.5

Product status: Stable — Studio feature; neutral profile from a 24-patch chart

Documentation validation: NOT REPLAYED

Evidence: audit of the SCSP v2 generator, refusals, and automated tests

Blocker: instrumented final-package generation and independent-corpus comparison not replayed

English mirror: NOT REPLAYED

Outcome: a generated .scsp profile selected on the active variant when the source is RAW, without changing the camera default

Calibration measures a ColorChecker 24-patch photograph before creative adjustments. Grading sliders are not embedded in this neutral CC24 profile.

Photograph the chart

- light it evenly;
- avoid specular reflections and cast shadows;
- expose without clipping bright patches;
- keep the complete chart large enough in frame;
- use the camera and lighting being calibrated;
- prefer RAW.

JPEG, TIFF, and PNG are readable but their color domain is less directly tied to RAW camera profiling.

Open the measurement image

Import the chart, select it alone in the Browser, wait for full Viewer display, then open **Calibration > Calibration — Color (CC24)**.

Choose the reference

Select the physical chart actually photographed:

- **ColorChecker Classic 24, GretagMacbeth 2005;**
- **ColorChecker Classic 24, X-Rite/Calibrite after November 2014;**
- **ColorChecker Classic 24, BabelColor average.**

0.6.5 limit — the illuminant selector shows D50 and D65, but generation uses D50 references and writes D50 to the profile. The EXIF-estimated D50/D65 suffix only names the file; it does not make D65 operational.

Place the four corners

1. click **Place Corners in Viewer**;
2. click the colored-area corners: top-left, top-right, bottom-right, bottom-left;
3. check **4 / 4** and ready state;
4. use **Clear** and repeat if order or position is wrong.

Enclose all 24 patches without cutting the chart. Measurement regions are placed inside the quadrilateral to avoid patch edges.

Generate the profile

Check **Profile File**, use **Browse...** if needed, click **Generate Profile**, then wait for used-patch count and mean/max residuals.

The proposed name uses camera make/model and EXIF-estimated illuminant. Output is strict .scsp v2. It declares post-white-balance camera RGB, linear working output, highlight headroom, illuminant, provenance, chart SHA-256, and LUT hash. Missing provenance, out-of-headroom content, or an incoherent contract is refused.

After generation from a RAW, ChromaLux installs a managed copy in the local camera-profile catalog, pins that exact artifact to the active variant, saves the selection in its sidecar, and rebuilds the Viewer. The chosen output file remains a separate export copy. The success message confirms local installation and selection. Generation never changes the camera default and never makes the profile eligible for automatic matching. Other variants, other RAW files from the same

model, and their thumbnails retain their own policy.

After checking independent exposures, scenes, and illuminants, you can use the separate action for setting a camera default. That promotion is a deliberate user decision distinct from measurement and does not replace existing explicit variant selections.

Check the result

Under **Adjustments > Base > RAW Interpretation**, verify that the profile appears in **Camera profile** and is selected on the active variant. Then compare the chart and independent real photographs from the same body. Inspect greys, skin, saturated colors, and highlights, keeping the previous profile until comparison finishes.

A low numerical residual alone does not guarantee preferable rendering on other scenes.

Return to another profile

Choose **Camera default — automatic fallback, Standard decoder camera matrix**, or another profile in **Camera profile**. This choice affects only the active variant. A source or export copy already installed in the managed catalog may be moved or deleted: the variant pins the managed copy. Only legacy loose SCSP profiles that were not installed in that catalog must remain at their path while referenced.

The separate, operational **Profile Editor** can derive a profile incorporating base curve, Exposure/Tonality, Color Editor, Color Balance, and Black & White. When working from a RAW, the saved derived profile is selected on the active variant after a managed copy is installed in the local catalog; the chosen file remains an export copy. Saving never promotes it to the camera default. It does not incorporate sharpening, Film Simulator, grain, or halation. This page covers neutral CC24 measurement only.

Historical SCSP v1 remains **legacy ambiguous**. Profile Editor does not promote it silently; derivatives remain v1 and no automatic migration occurs. Do not convert an arbitrary device ICC into SCSP to bypass the domain contract. See [Color management](#).

Annotate a Brief

Documented version: ChromaLux 0.6.5

Product status: Stable

Documentation validation: NOT REPLAYED

Evidence: audit of code, translations, and output formats

Blocker: annotation, persistence, and outputs not replayed in the final package

English mirror: NOT REPLAYED

Outcome: revisable visual instructions attached to a photograph and exportable with its development

Brief draws on the preview, describes zones, and attaches reference images.

Annotations remain separate from RAW development.

Before you begin

Select one photograph, wait for full display, then open **Brief** or the Brief tab under **Adjustments**.

Dictation also requires macOS microphone permission, a downloaded voice model, and the correct French or English note language. Typing, drawing, and export do not require them.

1. Draw and describe a zone

Enable **Show Brief Layer**, choose a tool, draw, enter the **Instruction**, choose Low/Normal/High/Critical priority, and optionally set **Zone Color**. Each numbered zone appears in the Viewer and **Instructions** list.

Shortcut	Tool	Use
V	Select / Move	Select, move, or resize.
R	Rectangle	Frame a rectangular area.

O	Oval	Circle a subject/detail.
P	Freehand	Draw a free outline.
B	Brush	Paint with opacity/hardness.
T	Text	Place text on the image.
E	Eraser	Erase a visual annotation.

These apply while Brief has focus. Drawing color and stroke width affect the next annotation; Brush also has opacity/hardness, Text has text size.

Warning — **Delete** removes both the selected zone and its instruction.

2. Organize annotation layers

The first annotation uses implicit **Annotations**. Click **Add** under **Brief Layers** for a full stack. Rename by double-click, reorder with arrows, set opacity, lock, or hide.

Warning — removing a layer also removes its instructions. ChromaLux asks for confirmation when it is not empty.

3. Add a general instruction and references

Use **General Brief** for image-wide intent or delivery constraints. Zones and the general brief can each have references: select the target, choose **Add Image / RAW...**, and inspect **References**. ChromaLux uses embedded RAW preview where possible.

Warning — ChromaLux stores the reference path, not a copy. Moving, renaming, or deleting the file can break preview and link.

4. Dictate an instruction

Prepare transcription

Open preferences with $\mathbb{A}$, choose **Brief — Voice Transcription**, select a model (Small balanced is recommended), choose French/English, download if needed, select input, and **Test Microphone**. First download requires Internet; larger models

need more storage and computation.

Record

Select a zone and click **Record**; click again or press Space to stop. Editable transcription appears under **Instruction**. General Brief has its own Record button. Optional auto-start begins after drawing a new non-Text zone.

Always proofread names and technical terms. Suggested priority is editable. Transcription is local; audio remains in memory and is not retained.

5. Restore after closing

Brief is automatically stored in <image-name>.cls.xmp: shapes, text, priority, layers, styles, reference paths, and general brief. Keep the sidecar with the original. Audio is never stored in it.

6. Export a Brief report

Export Brief Report... targets the current photograph:

Format	Output
PDF	Paginated developed image, visible annotations, instructions, priorities, general brief, available references.
PNG	Flattened developed image with visible annotations.
Markdown	Text instructions and local reference links.
JSON	Complete Brief structure.

The proposed name ends `_brief.pdf`; choose another extension when saving. Export requires a loaded photograph with Brief data. Hidden layers are not drawn in visual outputs.

7. Include Brief in a photo export

Check **Include Brief** in an export recipe:

- JPEG/PNG flatten visible annotations;

- **PSD (layered Brief)** retains background plus each visible Brief layer in an 8-bit PSD;
- TIFF writes composite, background, and each visible layer as named pages.

A cmu: derivation sidecar records the output. Report export in the Export panel is the same individual report.

Tip — hide annotations not meant for delivery and inspect the exported file. Brief reports are per current photograph, not a batch-review system.

Troubleshooting

- Missing zone: show Brief layer and check layer visibility/opacity.
- Zone will not move: choose **Select / Move** and unlock its layer.
- Dictation will not start: download model, grant permission, test microphone.
- Missing reference: restore its old path or re-add from the new path.
- Brief lost after move: restore `.cls.xmp` beside the original.

See [Develop a RAW](#) and [Sessions and files](#).

Create a contact sheet

Documented version: ChromaLux 0.6.5

Product status: Stable

Documentation validation: NOT REPLAYED

Evidence: audit of code, translations, and writers

Blocker: outputs not replayed in their target applications

English mirror: NOT REPLAYED

Outcome: a saved sheet exported as multilayer TIFF, JPEG, or both

Contact Sheet composes a Browser selection in an independent document. Up to four sheet windows can receive different selections.

Prepare the selection

Filter/select photographs in **Library**, check order and development, enter session information used in captions, then open **Contact Sheet**.

Create or choose a sheet

Open a new contact-sheet window or use toolbar **CS 1** through **CS 4**. Under **Open Sheets**, choose the destination sheet. Use **ChromaLux Native** for an independent document.

Adobe Photoshop is available only when detected and manages a PSB through its own workflow; native export below does not apply.

Set the document

Under **Sheet Settings**, choose preset/custom size, units, resolution, bit depth, ICC profile, cell size/type, and optional cell mask. Presets include A4, A3, 4K, 8K, press, and magazine cells. Check final pixel dimensions.

Build the grid

Set rows/columns, spacing, margins, colors, **Fill**, **Fit**, or **No Resize**, automatic extension, horizontal-photo handling, guides, snapping, and automatic placement.

Fill may crop to fill; **Fit** preserves the whole image and can leave space.

Add logo and caption

Choose a logo if needed; enter caption text and insert session/production labels; reorder labels; set font, size, alignment, color; optionally show file names. Empty session labels remain empty. Check preview.

Create and import

Click **Create Sheet**, change Browser selection if needed, then **Import Images** or ⌘B. Use **Update Sheet** after structural changes. With several sheets open, check which is active before the global shortcut.

Save and export

Save Document writes editable .contact-sheet.json.

Choose **Multilayer TIFF**, **JPEG**, or both under **Output Settings**, set folder/name, click **Export...**, and wait for progress. Photoshop .board references a PSB saved by Photoshop; native export does not manage it.

Final check

Check order, count, crop, names, and caption. Open TIFF/JPEG in an ICC-aware application and keep the sheet document with the session while it must remain editable.

Prepare a print

Documented version: ChromaLux 0.6.5

Product status: Stable; result depends on driver and paper

Documentation validation: NOT REPLAYED

Evidence: audit of code, translations, conversion, and soft-proof tests

Blocker: driver, profile, paper, lighting, and physical print not replayed

English mirror: NOT REPLAYED

Outcome: a print recipe, checked preview, and printed or exported output

Print composes a physical page from the focused photograph or selection and can send it to a driver or produce TIFF, JPEG, or PNG.

Select photographs

Focus one photograph for a single image. For a grid, select photographs in the desired order before opening **Print**. Contact-sheet mode places one selected image per cell; extra cells remain empty. Repeated-grid mode repeats the focused photograph.

Create a recipe

Load an existing **Recipe**, or enter a distinct name, set page/color/rendering, then **Save Recipe**. Recipes are associated with the image and stored in nearby ChromaLux data. Name them after paper, format, or lab.

Choose printer and paper

Under **Printer & Paper**, choose **Print to File** or **Printer**, page size or custom dimensions, orientation, and margins. Link margins when equal. Confirm they also respect the real device's non-printable area.

Configure color management

ChromaLux manages color

Choose application management, the exact printer–paper–ink ICC, perceptual or relative intent, and black-point compensation when required. Disable printer driver color correction.

Conversion starts from the actually rendered source profile: sRGB, Adobe RGB, or Display P3. ChromaLux does not force an intermediate sRGB conversion.

Printer manages color

Choose printer management and set profile/paper type in the driver. Do not apply the equivalent conversion in ChromaLux.

Warning — double color management often produces casts, wrong contrast, or oversaturation.

Use soft proofing

Enable **Soft Proof** to simulate the print profile. Paper simulation changes white appearance; gamut warning identifies difficult colors.

Simulation and gamut warning are independent. Display either, both, or neither. The warning changes neither simulated pixels nor output.

Main Viewer and Print use the same proof contract: real source profile, proof profile, intent, black-point compensation, and paper simulation. Do not compensate an unexpected discrepancy by arbitrarily choosing another profile.

Proofing does not replace a calibrated display, correct profile, or test print.

Build the page

Choose **Single Image**, **Grid — Repeat Image**, or **Contact Sheet**. For grids, set rows, columns, and gutters. Use **Fit** to preserve the image or **Fill** to crop into the cell; anchors select the retained area.

Set resolution, grain, and sharpening

- **Target PPI** controls page density.
- **Output Bit Depth** offers 8 or 16 bits depending on format.

- **Output Sharpening** compensates final reproduction; excess radius halos.
- **Print Grain** is sized in physical paper units.

Judge grain and sharpening at **100% — Print Pixels**, not only fit view.

Add a caption

Caption can show file name, exposure, rating, and capture date. Set font/size and check that it does not consume excessive image area.

Check preview

Open print preview, inspect **Fit**, enable centimetre rulers, switch to **100% — Print Pixels** for texture/detail, then adjust the recipe; preview updates.

Print or export

In preview:

- **Print...** opens the macOS print dialog;
- **Export Image...** creates TIFF, JPEG, or PNG.

TIFF can be 8/16-bit. A CMYK profile forces TIFF; JPEG/PNG are RGB. Check the final message: if full-resolution rendering fails, ChromaLux may report use of the preview as source.

Before delivery, verify dimensions, declared resolution, bit depth, and embedded profile. For direct printing, ensure the driver does not repeat ChromaLux color management.

See [Color management](#).

Activities, tabs, and modules

- Documented version:** ChromaLux 0.6.5
- Product status:** Stable, with incomplete features identified
- Documentation validation:** NOT REPLAYED
- Evidence:** audit of activity/module registries and translations
- Blocker:** layouts and navigation not replayed in the final package
- English mirror:** NOT REPLAYED
- Outcome:** organize tools without changing photographs or processing order

Three levels

Level	Example	Role
Activity	Adjustments	Complete work stage.
Module tab	Base, Colors	Organizes modules in an activity.
Module	Exposure, White Balance	One tool or setting group.

The left Activity Bar changes activity; module tabs appear above the module column. The selected photograph remains the same.

Warning — visual module order does not change processing-graph order. Moving Film Grain above Film Curve does not calculate grain first.

Activities delivered in 0.6.5

Activity	Main delivered modules	Status or limit
Library	Finder, Filters , Browser at right	All editions.
Calibration	CC24, Profile Editor,	Studio; last three announce

	Scientific, Detector, Geometric	future phases.
Tethered Capture	capture, next name, folder, camera controls, session details	Studio; automatic Nikon/Canon/Fujifilm, explicit gPhoto2 fallback.
Adjustments	base, WB, exposure, extended dynamic range, detail, primaries, curves, levels, color, lens, layers/masks	Main development/grading activity.
Film Simulator	Black & White, Film Curve, Film Grain, Halation, Darkroom Print, Vignette	Functional; availability can depend on image context.
Science	Viewer overlays, false color, channels, line profile, Astro/Microscope, PT contrast, measurement	Specialized; some depend on format/context.
Contact Sheet	Contact Sheet	Default-on, can be hidden by configuration.
Metadata	session details, keywords, photo metadata, AI metadata	Local AI depends on Ollama and a compatible vision model.
Brief	Brief	Default-on, can be hidden.
Export	Export	Writes checked recipes.
Print	Print	Printed output; distinct from creative Darkroom Print.

Default **Adjustments** tabs:

- **Base:** RAW Interpretation, White Balance, Exposure, Extended Dynamic

Range, Detail, Primaries;

- **Adjustments:** SCS Curves/Levels, Tone Equalizer, Dehaze, Fringes, Lens Correction, Vignetting;
- **Colors:** Color Balance, Color Editor, Harmony;
- **Layers & Masks;**
- **MockUp:** Raw Mockup and ChromaKey;
- **Brief**, when enabled.

Histogram, SCS Parade, and SCS Vectorscope sit separately above modules. **Lens Correction** is a module; **Perspective** is a distinct toolbar tool.

Identify incomplete features

- Calibration **Scientific**, **Detector**, and **Geometric** are placeholders.
- **Inspector** and **Film** are registered as future panels.
- Legacy future **Tether** and **Style** panels do not replace current tether or presets.
- **Node Editor v1** is read-only.
- Local grading supports Fisher Curve, Exposure, White Balance, Color Editor, Color Balance, Primaries, Black & White, and Extended Dynamic Range with chromatic/SCS/gradient/painted masks. AI depends on Apple Vision or installed SAM. Clone/Heal are disabled.
- **Extensions** still opens a future-version interface.

Experimental feature — do not treat these future panels as guaranteed production workflow elements.

Add a module

Open the destination activity/tab, click **Add Module** at the column bottom, then choose from the category. The menu can also create a new tab or a tab in Column 2.

Tools opens the same catalogue but normally creates a floating panel near the pointer. Multiple instances are allowed; keep one instance of each image-changing module for predictable work.

Module headers

Depending on capability:

- power bypasses/reactivates processing;
- chevron collapses/expands;
- ... opens save/apply preset, copy/paste module, and reset.

Copy is temporary and same-module-type only. Save a preset for durable reuse.

Reorder or move modules

Drag headers within a tab, to another tab, or to the other column. The context menu can move up/down, remove from tab, or move to a named/new tab.

Removing a module does not delete saved photograph adjustments. Re-adding it restores focused-photo values.

Organize tabs and columns

Tab + creates a tab. Its context menu adds, renames, deletes, reorders, or moves between columns.

Warning — Delete Tab immediately destroys its panels without confirmation.

Photograph adjustments remain, but that tab layout is lost.

Create a custom activity

Activity Bar + creates an empty custom activity. Name it, then add tabs and modules. Only custom activities can be renamed/deleted; deletion confirms and removes that workspace.

Save a complete layout

Save all activity layouts with **View > Workspace Layout > Save Current Layout As....**

Warning — applying a layout preset closes and relaunches ChromaLux. Save work first.

Return to the delivered layout

View > Reset Default Layout clears stored UI state, including custom activities, but not originals, metadata, adjustments, or separately saved layout presets. A visible custom activity can persist until next launch.

For more information

See [Tour the interface](#), [Layers and masks](#), and [Film Simulator](#).

Keyboard shortcuts

Documented version: ChromaLux 0.6.5

Product status: Stable

Documentation validation: NOT REPLAYED

Evidence: audit of shortcut registry, keyboard handlers, and translations

Blocker: complete keyboard workflow and VoiceOver not replayed in the final package

English mirror: NOT REPLAYED

Scope: shortcuts registered by default in the macOS application

Customize under **View > Keyboard Shortcuts....** Saved customization replaces this page's defaults. On macOS, ⌘ is Command, ⌥ Option, ⌘ Control, and ⇧ Shift.

Sessions and general commands

Shortcut	Action
⌘,	Open preferences.
⌘N	Create a session.
⌘O	Open a session.
⇧⌘T	Reopen last session.
⌘I	Import images.
⌘Z	Undo.
⇧⌘Z	Redo.
⌘R	Reset all adjustments on the focused photo.
⌘⌥A	Show/hide Activity Bar.

Warning — ⌘R resets the complete development. Use a module power button to bypass only that module.

Browser and selection

These apply when Browser/thumbnails have focus and do not intercept text-field typing.

Shortcut	Action
←, ↑	Focus previous photo; in multiple selection, retain selection.
→, ↓	Focus next photo; in multiple selection, retain selection.
⇧ + arrow	Extend selection from anchor.
⌘A	Select all currently visible after filtering.
0	Clear rating.
1 through 5	Assign one to five stars.
P	Pick selection.
X	Reject selection.
U	Clear pick/reject.
=	Enlarge thumbnails.
–	Reduce thumbnails.
⌘ or Delete	Request move to session Trash; confirmation shown.

Color labels

Labels are cumulative toggles; ⌘0 clears all.

Shortcut	Label
⌘0	None

⌘1	Grey
⌘2	Green
⌘3	Purple
⌘4	Blue
⌘5	Yellow
⌘6	Red
⌘7	Orange

Viewer and display

Shortcut	Action
←, ↑	Previous photograph.
→, ↓	Next photograph.
I	Show/hide Viewer Info Bar.
/	Show rendered RAW without user adjustments, then return.
J	Toggle output clipping warning.
⇧J	Toggle pre-recovery sensor clipping by channel.
F	Toggle SCS false color.
\	Bypass Science analysis overlays.
M	Show/hide selected editable-mask tool.
⇧M	Cycle Image, Overlay, Black & White mask display.
Escape	Cancel picker, measurement, or temporary tool.
⌘1 through ⌘9	Focus corresponding comparison tile.

/ compares development; \ controls Science overlays. Mask shortcuts require a selected editable mask.

Interface navigation

Shortcut	Action
Tab / ⌘Tab	Next / previous exposed focus target.
←	Collapse focused module header.
→	Expand focused module header.
Space or Return	Activate focused button/checkbox.
⌘F10 or Menu key	Open module-header context menu when available.
Return	Apply active Perspective guides.
Escape	Cancel Perspective editing or temporary tool.

Perspective endpoint movement is **PARTIAL** by keyboard; a pointer is needed.

VoiceOver is **NOT REPLAYED**. See [Accessibility](#).

Development and setting transfer

Shortcut	Action
⌘C	Copy all adjustments from focused photo.
⌘V	Paste copied adjustments to selection; toolbar menu limits groups.
C	Enter/leave Crop.
H	Enter/leave Hand tool.
O	Cycle crop composition guides.
⌘X	Swap crop-frame orientation.
L	Arm horizon-line drawing.

⇧F	Detect/frame the last Focus target type.
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Second Viewer, Focus Loupe, and next Focus face are configurable but have no 0.6.5 default shortcut. Every slider exposes configurable increment/decrement actions, also with no defaults.

Metadata and contact sheet

Shortcut	Action
⇧⌘↵	Analyze selection with AI metadata and auto-accept valid proposals.
⇧⌘B	Add Browser selection to current contact sheet.

AI auto-accept requires ready Ollama/vision model and writes valid proposals without review. Contact-sheet windows have configurable but unassigned shortcuts.

Double Exposure workspace

Shortcut	Action
A	Show layer A.
B	Show layer B.
⇧C	Show composite.
R	Reset assembly.
⌘↵	Calculate and save composite.

Resolve conflicts

Open **View > Keyboard Shortcuts...**, locate both actions, change one or restore its default, then focus Viewer or Browser as required. Single-letter shortcuts do not fire inside text fields.

Accessibility and assisted navigation

Documented version: ChromaLux 0.6.5

Product status: structural support on primary workflows; incomplete coverage of graphical tools

Documentation validation: NOT REPLAYED

Evidence: audit of accessible names, descriptions, tab order, target sizes, and automated widget tests

Blocker: VoiceOver NOT REPLAYED; keyboard movement of Perspective handles PARTIAL

English mirror: NOT REPLAYED

Outcome: navigate primary areas by keyboard and identify limits that still require a pointer

ChromaLux exposes accessible names and descriptions on primary import, Browser, layer, mask, and export controls. Code and widget tests cover these properties; the complete workflow has not been replayed with VoiceOver in the final package.

VoiceOver status — NOT REPLAYED. This page is not a VoiceOver certification. If production depends on a screen reader, first replay the procedures with non-confidential files and record any defects.

Set density and contrast

Open **ChromaLux > Preferences... > General > Appearance**.

- **Comfortable** shows Activity Bar labels and uses roomier targets.
- **Compact** narrows the bar and uses icons; activity names remain exposed to assistive technologies and tooltips.
- **Neutral color environment** removes decorative color from the judgment environment.
- A custom palette is refused if text does not reach 4.5:1 minimum contrast

against every surface.

Start with **Comfortable** for visual or motor navigation. Density changes apply without a restart in adaptive areas.

⌘A shows or hides the Activity Bar without changing the current activity.

Import by keyboard

File > Import Images... has a stable tab order:

1. source;
2. destination;
3. backup, naming, metadata, adjustments, filtering, and post-import actions;
4. review-grid selection;
5. **Cancel**, then **Import**.

Use Tab and ⇧Tab to move. Labels are associated with fields and selection buttons have a minimum 28 × 28 pixel target.

Status distinguishes:

- source scan in progress;
- no importable image;
- no filter result;
- selected image count;
- resolved destination;
- the first reason import is blocked.

Import remains disabled until scanning ends, at least one importable image is selected, and the destination is valid. Its accessible description repeats the blocking reason.

Below 980 pixels wide, the window becomes one vertical column. After a display change, its position is clamped to the visible area.

Understand the Browser

The Browser separates:

- the **selection** targeted by batch commands;

- the **focused photo** feeding panels and several commands;
- the **photo displayed in the Viewer**, which can temporarily differ.

Its accessible description announces the selected count, focused file name, whether it belongs to the selection, and the displayed file name when different.

With focus in the Browser:

- ←, →, ↑, and ↓ move the focused photograph;
- ⌘ + arrow extends selection;
- ⌘A selects visible images;
- rating, status, and color commands are listed in [Keyboard shortcuts](#).

In a comparison grid, ⌘1 through ⌘9 focus the corresponding tile.

Navigate modules, layers, and masks

Module headers expose title and expanded/collapsed state. With a header focused:

- ← collapses;
- → expands;
- Space or Return activates the focused control;
- Menu key or ⌘F10 opens its context menu when available.

The **Layers** panel exposes stack, add/remove, opacity, blend mode, and tool display. + and – buttons have a 28 × 28 pixel target.

Each mask announces type, enabled state, and expanded/collapsed controls. Tab reaches exposed fields and buttons; Space activates a focused checkbox or button. Painting gestures, local points, and gradient handles still require the Viewer and a pointer.

Use Perspective

Perspective opens a palette whose **Vertical**, **Horizontal**, **Complete**, **Apply**, **Cancel**, and **Reset** controls are reachable by tabbing.

- Return applies active guides.
- Escape cancels the current edit.

Perspective keyboard status — PARTIAL. Guide endpoints cannot yet be fully

selected and moved by keyboard. Accurate placement requires a pointer; Return and Escape alone do not make the complete workflow accessible.

Export and follow progress

In **Export**, check a recipe, then use **Export Current Image** from the empty queue state. If no export has completed, **Open Destination Folder** opens the configured destination directly.

The progress bar exposes an accessible name and updated description. Pause, resume, and stop keep their scope on the current queue. A background export does not make it safe to quit before checking the result.

Known limits and replay protocol

The packaged application still requires replay for:

- actual VoiceOver announcement order in every activity;
- spoken feedback after selection/focus changes;
- nested context-menu navigation;
- spoken progress during long imports and exports;
- focus after floating windows open or close;
- effective contrast with macOS profiles and accessibility settings;
- graphical Viewer tools beyond their command buttons.

When reporting a defect, record activity, announced control, expected and actual announcement, navigation method, and macOS version. Exclude client names and personal paths from captures and recordings.

Formats and compatibility

Documented version: ChromaLux 0.6.5

Product status: Stable; compatibility depends on format and camera

Documentation validation: NOT REPLAYED

Evidence: audit of import, indexing, decoding, ICC management, and export paths

Blocker: hardware matrix and private corpus unavailable for complete replay

English mirror: NOT REPLAYED

Principle: detection during import does not imply development in Viewer

Compatibility has four levels: import detection, Browser visibility, catalogue indexing, and Viewer decoding. Their 0.6.5 lists differ.

Requirements

ChromaLux 0.6.5 targets Apple Silicon Macs running macOS 26 Tahoe or later. It is not universal and does not target Intel Macs. Photo-format support does not establish tether compatibility.

Formats detected by Import Images

Case-insensitive extensions:

- Nikon: NEF, NRW;
- Canon: CR2, CR3, CRW;
- Sony: ARW, SR2, SRF;
- Fujifilm: RAF;
- Panasonic: RW2;
- Olympus/OM System: ORF;
- Pentax: PEF;
- generic/Adobe: DNG;
- Phase One/Hasselblad: IIQ, 3FR;

- Sigma: X3F;
- GoPro: GPR;
- rendered: JPG, JPEG, HEIC, HEIF.

Detection means the file can be proposed/copied, not necessarily shown or developed. X3F/Foveon is bundled, but encoding and geometry are checked before development and can be explicitly refused.

Browser visibility and catalogue indexing

Family	Browser-visible	Indexed
Viewer-routed RAW	NEF, RAF, CR2, CR3, DNG, ARW, IIQ, X3F	Yes
Common images	JPG, JPEG, PNG, TIF, TIFF, HEIC, HEIF, WEBP	Yes
Scientific images	FITS, FIT, FTS	Yes

PNG, TIFF, WebP, or FITS manually placed in **Captures** can appear although the importer does not propose them. Conversely, RW2, ORF, PEF, NRW, CRW, SR2, SRF, 3FR, and GPR can be copied by import but not recognized later by the 0.6.5 Browser.

Warning — check thumbnails before erasing or formatting a card. A successful copy is not proof of development support.

Viewer-developable formats

RAW development

Extension	Route
NEF	Bayer RAW according to identified sensor
CR2, CR3	Bayer RAW
ARW	Bayer RAW
DNG	Bayer RAW

RAF	X-Trans RAW
IIQ	Phase One Bayer route
X3F	dedicated Foveon route when encoding/geometry are supported

These are routes, not camera certification. Compression variants, sensors, or third-party DNG can still fail.

Experimental feature — the HALO X-Trans RAF route remains experimental in 0.6.5. Qualify body and compression mode before production.

X3F accepts seven validated encoding families and refuses unknown encoding, SDQH2, or incoherent layer geometry. Some ChromaLux-created 16-bit TIFFs use a .dng extension; ChromaLux recognizes them as rendered rather than sensor RAW.

Rendered images

JPG, JPEG, PNG, TIF, TIFF, HEIC, HEIF, and WEBP use the rendered-image loader. HEIC/HEIF/WebP also depend on packaged/macOS decoders. They do not have RAW highlight headroom or sensor data.

In color pipeline v2:

- embedded ICC is read and converted to the linear working domain;
- unprofiled JPEG/PNG/TIFF/HEIC/HEIF/WebP is explicitly assumed sRGB;
- a ChromaLux rendered DNG identified as linear retains its linear contract;
- pixels travel at high precision without a forced RGBA8 round-trip.

Historical v1 variants retain legacy behavior and are not implicitly migrated.

FITS

FITS, FIT, and FTS use the scientific loader and catalogue. Import Images does not propose them. Put a working copy in **Captures**, open it in the Browser, and keep the scientific original elsewhere.

Export formats

Format	Depth	Main options	Common use
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JPEG .jpg	8-bit	quality 1–100	lightweight proof, Web, sharing
TIFF .tif	8/16-bit	none, LZW, ZIP	master, retouching, print
PNG .png	8/16-bit	output depth	review, graphics
PSD .psd	8-bit	separate Brief layers	annotation handoff to validate in target

TIFF/PSD writers are designed to preserve Brief pages/layers, but target-reader interoperability still requires replay. JPEG/PNG flatten annotations when **Include Brief** is enabled.

Export ICC choices are sRGB IEC61966-2.1, Adobe RGB (1998), and ProPhoto RGB. The recipe **ICC** is a delivery profile and differs from **Viewer render space** (sRGB, Adobe RGB, Display P3). Export converts from the actual render space and embeds the recipe profile; without explicit replacement, it preserves the rendered-space profile.

Never choose the display ColorSync profile as delivery ICC merely because it has the display name.

Resolution writes PPI/PPCM metadata; **Scale** changes dimensions. **Never Upscale** prevents output larger than source rendering.

Special outputs

- Contact Sheet exports flattened JPEG, paged TIFF, or both.
- Brief reports export PDF, annotated PNG, Markdown, or JSON separately.
- Internal JXL previews are not a 0.6.5 delivery format.

Associated files

- .clsession: session catalogue;
- .xmp: descriptive metadata;
- .cls.xmp: adjustments, variants, ChromaLux data;
- .scsp: camera input profile; v1 is **legacy ambiguous**, v2 declares domains, headroom, provenance, and hashes;

- `.jxl` under ChromaLux folders: regenerable previews/caches.

Qualify a new body or format

1. import a non-confidential example with production RAW settings;
2. check Library visibility;
3. wait for complete Viewer development;
4. inspect orientation, dimensions, color, highlights, metadata;
5. change Exposure and White Balance slightly;
6. quit/reopen to check persistence;
7. export sRGB JPEG and 16-bit TIFF and open them in an ICC-aware application.

An embedded JPEG thumbnail does not prove sensor decoding. See [Troubleshooting](#) and [Color management](#).

Scopes and the Science activity

Documented version: ChromaLux 0.6.5

Product status: Stable scopes; identified Science functions Experimental

Documentation validation: NOT REPLAYED

Evidence: code, translation, and automated-test audit

Blocker: missing reference images and final-package perceptual replay

English mirror: NOT REPLAYED

Outcome: inspect light, color, and measurements without changing development

Scopes analyze the Viewer image; **Science** adds diagnostic maps, guides, and measurements. They do not correct the photograph or change RAW Interpretation.

Show a scope

Adjustments can display Histogram, SCS Vectorscope, and SCS Parade. Add missing ones through **Tools > Analysis** or **Add Module**. They follow the focused photograph and current render and can show RAW decoding progress.

Histogram

Left-edge mass indicates many dark tones; right-edge mass indicates bright tones, not automatically bad exposure.

- **EV** shows available pre-render-curve scene data for RAW.
- **Adobe**, **P3**, or **sRGB** shows RGB/luminance in active output domain.

Rendered files use output domain. Hover for RGB, luminance, SCS, and available scene luminance. Use J for developed-output clipping and ⌈J for pre-recovery sensor channels.

SCS Vectorscope

Center is neutral/low saturation; radius is saturation; direction is hue family.

Temperature/tint axes help identify global casts. A distant trace can be intentional vivid color.

Zoom changes reading scale. Options include tonal-range separation, skin-tone axis/tolerance, saturation alert, rings/axes/diagonals/color markers, colored/monochrome background, linear trace, and smoothing.

Right-click sets a target; ⬆ + right-click or clear removes targets.

Warning — Vectorscope describes displayed signal. It proves neither print gamut conformance nor camera calibration.

SCS Parade

Parade retains horizontal image position.

Mode	Reading
RGBL+Sat	RGB, luminance, saturation
RGB	three color channels
L	luminance only
Sat	saturation normalized to SCS limit
SCS	chromatic temperature/tint directions

Gain enlarges only the trace. Options can use RAW scene EV, overlay saturation, choose Legal 0–1/Rec.709/P3/Rec.2020 gamut targets, qualify a range, retain reference trace, or export PNG/CSV. Alerts identify values; they do not correct them or replace ICC soft proofing.

Science modules

Module	0.6.5 function
Viewer Overlays	Grid, crosshair, scale bar, line-profile guide, pixel/unit conversion.
False Color	Temporary luminance, temperature, or saturation

	map.
Channel Analysis	RGB, approximate SCS chromatic reading, or greyscale luminance.
Line Profile	Intensity/chromaticity view and fixed diagonal guide; profile function incomplete.
Astro / Microscope	Two previews and FITS rotation; announced processing remains limited.
PT Contrast (ICP)	Experimental modular residual, entropy, local divergence, and persistence maps.
Measurement & Annotation	Distance, area, angle, persistent label.

Only one analysis map is active. These maps change Viewer presentation only and are not ordinary exports.

Configure guides and scale

Enable overlays; choose measurement grid, center crosshair, calibrated scale bar, or line guide; set opacity; enter **Units / Pixel** and unit label such as mm or μm ; set grid step or 0 for ten automatic divisions.

ChromaLux does not infer this geometric calibration from the subject.

Press \ to bypass Science maps, guides, and measurements temporarily.

Warning — this does not bypass development or Brief annotations. Use / to compare the current development with rendered RAW without user adjustments.

Measure and annotate

After setting units:

1. **Line**: two clicks for distance;
2. **Rect**: two corners for area;
3. **Angle**: three points, second is vertex;
4. **Text**: click and enter label;

5. **Done / Pan (Escape)** returns clicks to Viewer panning.

Double-click a table label to edit. Select a row + Delete for one measurement. **Show in Viewer** toggles all; **Clear All** deletes all current-photo measurements. They persist per image in `.cls.xmp`.

Experimental limits

- **Readout Only** currently produces the same map as Viewer Overlay.
- **Line Profile** has no sample curve or movable line; its guide is fixed.
- Astro **Background Model** is generic contrast, not astronomical background subtraction; **Flat Field** is a monochrome preview, not flat-field correction.
- FITS rotation affects FITS only.
- Some planned options are not exposed.
- **PT Contrast (ICP)** is neither a standard photographic measurement nor a grading effect.

What is retained

Measurements persist. Some scope preferences persist. Active maps, guide calibration, and several Science settings reset on restart. Analysis is never written to the original.

Initial troubleshooting

Documented version: ChromaLux 0.6.5

Product status: Stable

Documentation validation: NOT REPLAYED

Evidence: audit of code, translations, and known regressions

Blocker: manual scenarios not replayed in the final package

English mirror: NOT REPLAYED

Principle: protect originals before attempting repair

Record the action that triggered the issue. Do not delete a session, sidecar, or cache before copying it.

Files to protect

For `image.nef`, protect the original, `image.nef.xmp`, `image.nef.cls.xmp`, `.cls.session`, internal `.cls`, and regenerable previews. Back up the complete session folder first.

ChromaLux will not launch

Check Apple Silicon, macOS 26 Tahoe or later, application copied to **Applications** rather than run from DMG, and expected published DMG version. Do not bypass unusual macOS security warnings through Terminal; retain the exact message and verify provenance.

Interface remains in English

Choose **View > Language > French**, quit fully, relaunch. Some technical names and acronyms intentionally remain English.

Session will not open

Check `.clsession` extension, mounted/writable volume, and open the file through **File > Open Session...**, not its parent folder. After moving a session, open the catalogue in its new location. Do not manually edit it.

Imported photographs do not appear

Open **Library**, choose the import destination in ChromaLux Finder, clear filters, confirm files in macOS Finder, and wait for indexing/thumbnails. If absent on disk, inspect the import report and retain the source card.

Viewer is blank

Click a supported thumbnail, wait for initial RAW decode, try another file and a known JPEG, check source path, reset default layout if Viewer is hidden, then close/reopen the session before touching sidecars.

0.6.5 RAW routes cover NEF, CR2, CR3, DNG, ARW, RAF, IIQ, X3F, but not every body/variant. Unsupported X3F encoding/geometry is explicitly refused. Import can copy other extensions that Viewer cannot develop.

Adjustments are not visible

If **BEFORE** is visible, press `/` or disable the RAW-without-adjustments command. Check focused variant/photo, module bypass, recent `⌘Z`, and associated adjustment file.

One adjustment affected several photographs

Synchronization is on by default. Stop adjusting, undo with `⌘Z` while available, reduce selection to one, and inspect several affected images.

Interface or scopes disappeared

Use **View > Reset Default Layout**. It restores layout without touching photos or adjustments. Save reusable personal layouts under **View > Workspace Layout**.

Display profile is unavailable or changes

Open **Preferences > Viewer > Color > Active display profile**. It follows the window's display. Wait after moving windows, check **System Settings > Displays > Color**, and inspect calibration date/report if rendering seems wrong. Never use display profile as delivery ICC just to make export “match.” The tooltip's SHA-256 can identify the read profile.

An SCSP profile is refused

Strict v2 rejects incoherent domains, headroom, provenance, hashes, or content. Do not hand-edit header/JSON. Keep the exact message, return to the previous profile, regenerate from Calibration, and compare on a new variant.

Historical v1 can remain **legacy ambiguous** and is never migrated by renaming.

Export button is disabled

A photo must be loaded and at least one recipe checked. Then check destination exists/is writable. A multiple selection queues every selected item for every checked recipe.

Export differs from Viewer

Check, in order:

1. checked recipe;
2. **Viewer render space**;
3. recipe ICC and format;
4. profile actually embedded;
5. soft-proof state;
6. viewing application's ICC handling;
7. dimensions and output sharpening.

Proofing simulates a destination but is not baked into production export. Gamut warning is independent. See [Color management](#).

Keyboard navigation or VoiceOver is incomplete

See [Accessibility](#). VoiceOver is **NOT REPLAYED** and Perspective handle movement

remains **PARTIAL**. An accessible name in code is not a successful reader replay.

Ollama or AI metadata does not work

It requires local Ollama and a vision-capable model. See the [Apple Vision + Ollama appendix](#). Review proposals; do not force metadata writes after model/sidecar errors.

Report an issue

Include ChromaLux version, macOS/Mac model, camera/extension, minimal steps, exact message, non-confidential capture, and whether one or all photographs are affected. Do not attach client RAW/session data without authorization.

Experimental appendix — Local Apple Vision + Ollama AI metadata

Documented version: ChromaLux 0.6.5

Product status: Experimental appendix — local proof of concept, Apple Silicon macOS

Validation inherited from the French page: code and test audit; complete final-package replay to be recorded

English mirror: NOT REPLAYED

Writing: human approval in normal workflow; automatic writing only after explicit auto-accept invocation

This optional local graph enriches **Metadata**:

analysis.preview

```
└─> ai.apple_vision ─┘
```

```
|-----> ai.ollama -> metadata.proposal
```

Apple Vision supplies classification/OCR clues. An Ollama vision model receives them and a bounded JPEG preview, then proposes title, description, keywords, and uncertainties. A proposal remains in the session until accepted/rejected.

Requirements

- Ollama installed and running locally;
- an Ollama model declaring `vision`;
- an open ChromaLUX session with at least one catalogued selection.

ChromaLux bundles/downloads no model weights. Suggested options:

ollama pull gemma3:4b

```
# or higher quality with more memory:
```

ollama pull gemma3:12b

The PoC checks `/api/show` for `vision`, and **Test / Refresh** must have observed the model digest through `/api/tags`. This makes local state auditable but does not cryptographically pin a mutable tag.

After `/api/chat`, ChromaLux checks `/api/ps`; loaded canonical name and digest must match the selection or the image fails without a reviewable proposal. This is local execution evidence, not remote signed attestation.

`/api/tags` provides digest while `/api/show` is authoritative for capability, because some Ollama versions omit `vision` from tags.

Opt-in developer smoke

The real macOS Apple Vision + Ollama smoke is disabled until `CLS_OLLAMA_LIVE_MODEL` is set. Fetch the unversioned Kodak fixture once:

```
./tools/fetch_kodak.sh
```

Variables:

- `CLS_OLLAMA_LIVE_MODEL`: required; untagged `gemma3` resolves only to `gemma3:latest`, then uses Ollama's exact canonical result;
- `CLS_OLLAMA_LIVE_ENDPOINT`: default `http://127.0.0.1:11434`;
- `CLS_OLLAMA_LIVE_IMAGE`: default `tests/fixtures/kodak/kodim04.png`.

```
CLS_OLLAMA_LIVE_MODEL=gemma3:4b \
```

```
CLS_OLLAMA_LIVE_ENDPOINT=http://127.0.0.1:11434 \
```

```
CLS_OLLAMA_LIVE_IMAGE="$PWD/tests/fixtures/kodak/kodim04.png" \
```

```
QT_QPA_PLATFORM=offscreen \
```

```
./build-work/bin/test_ollama_metadata_client \
```

```
liveAppleVisionOllamaSmokeIsOptIn
```

The smoke reads tags before/after inference and checks loaded weights through `/api/ps`. It fails on tag change or identity mismatch.

Use the feature

1. launch Ollama and open a session;

2. select Browser photographs;
3. open **Metadata > AI metadata — Apple Vision + Ollama**;
4. keep `http://127.0.0.1:11434`;
5. click **Test / Refresh** and choose an installed vision model;
6. choose output language;
7. click **Analyze N photo(s)**;
8. review/edit title, description, keywords, and uncertainties;
9. click **Accept**, **Reject**, or **Accept All Valid**.

0.6.5 offers French, English, German, Spanish, and Italian.

The batch is an immutable snapshot of the selection at Analyze time. **Cancel** stops current work and retains completed proposals.

Start Ollama automatically

Under **Preferences > Metadata**, enable startup/preload for the next ChromaLux launch after verifying the selected model. ChromaLux can start installed Ollama and preload downloaded weights; it neither installs Ollama nor downloads models. Quitting ChromaLux does not unload the model.

The photographic prompt is customizable; ChromaLux retains language, evidence, and structured-output constraints.

Auto-accept

Global  analyzes selection and accepts every valid proposal. It is configurable.

Warning — this skips review. Use normal Analyze when metadata must be checked before writing.

Writes after acceptance

Acceptance:

- merges proposed and existing XMP keywords without case-insensitive duplicates and without deleting existing keywords;
- replaces title/description only when proposal is non-empty;
- preserves foreign XMP namespaces;

- synchronizes SQLite mirror and session search index.

Malformed existing XMP blocks acceptance and stays intact. Rejection never touches XMP. Standard `<image>.xmp` is written beside the original. ChromaLux prepares SQLite, atomically replaces XMP, then commits; normal failure restores the previous sidecar.

Privacy and safeguards

- default endpoint is restricted to `127.0.0.1`, `localhost`, or `:::1`;
- Ollama requests bypass system proxies;
- no cloud service is used;
- image bytes, prompts, and complete responses are not logged;
- OCR/Vision labels are untrusted data, never instructions;
- Ollama response must satisfy strict JSON Schema;
- normal workflow keeps human approval; auto-accept is a distinct explicit action.

Persistence and reuse

Each run records image/variant fingerprint, graph/prompt versions, Apple Vision revisions, endpoint, and Ollama model/digest. The fingerprint also covers source size/date, preview recipe, and visual `.cls.xmp` content.

A pending proposal is reused only when all match. Successful new analysis makes older pending proposals obsolete while preserving provenance. Fingerprint is checked again before acceptance; visual changes cause refusal and require reanalyzing.

PoC limits

- Ollama runs sequentially for predictable memory.
- Output is model-dependent, not factual truth or calibrated confidence.
- Apple Vision exists on macOS; other-platform stub reports unavailable clues.
- Graph is serializable/visible but not editable in general Node Editor.
- Language quality depends on model.

- XMP and SQLite are separate resources: ordinary errors are compensated, but a crash exactly between XMP replacement and SQLite commit can require reconciliation from portable-source XMP.

Quick diagnostics

- Ollama unavailable: `curl http://127.0.0.1:11434/api/version`.
- No model: install `gemma3:4b` or `gemma3:12b`, then **Test / Refresh**.
- Missing digest: select an installed-list model rather than hand-entering tag.
- No vision: choose a model whose `/api/show` includes `vision`.
- `vision` missing from tags: not blocking if `/api/show` declares it.
- Invalid response: use a recent structured-output-capable model; ChromaLux does not heuristically repair JSON.
- Image changed: analyze again.
- XMP refused: repair/back up existing sidecar; it is not silently overwritten.

References: [Ollama Vision](#), [Structured Outputs](#), [Chat API](#), and [running models / api/ps](#).